

hampers $\rightarrow$ Σ state

eg: The swimmer was hampered by his clothes.

Note:

It's complement of subtracted. (ie 00110110) $\rightarrow$

~~10001001~~

11001001

$$y = \bar{A}A + A\bar{A}$$

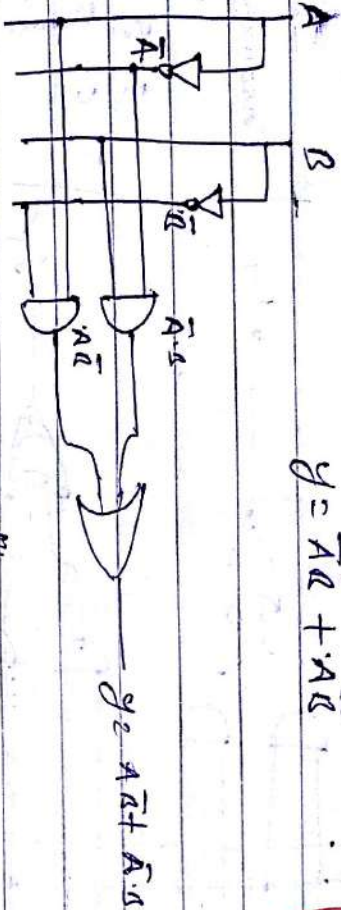
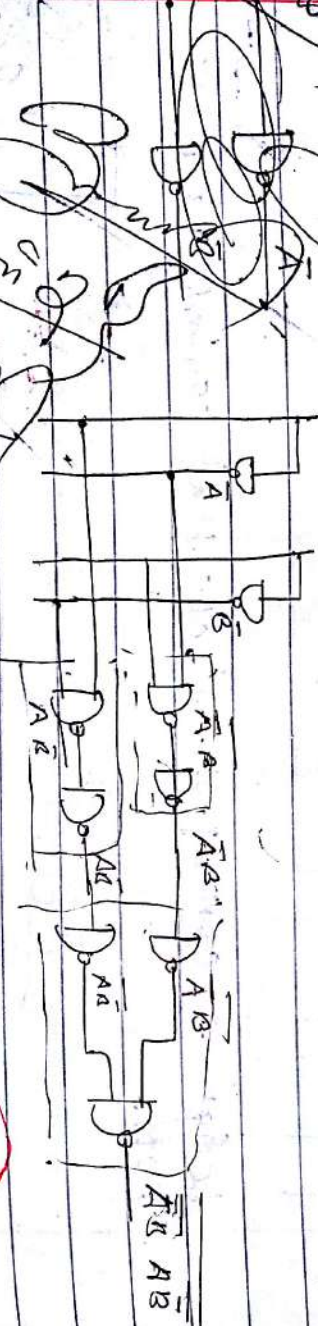
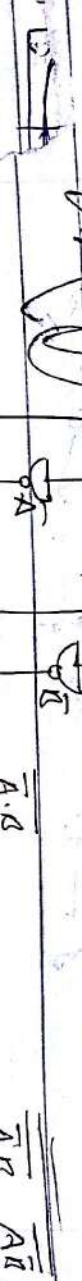


Fig: Ex-or ckt using basic gate.

$$A\bar{B} + \bar{A}B$$



final design is



Mantba Toshi
→

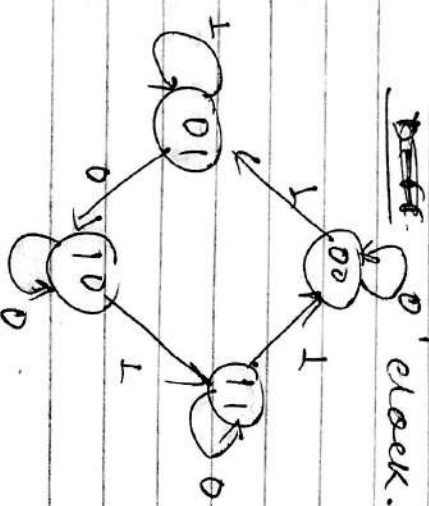
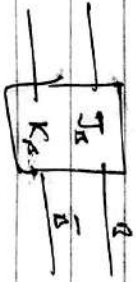
5. c/c

1 bit of a time

510221g

Feed had one
more corn.

base on clock $\frac{J-K}{2}$ updated based on



BCD counter

~~FF~~ $\bar{Q}_4$

Count sequence

~~FF~~ $\bar{Q}_4$

O/P

Q_3 Q_4 Q_2 Q_1 TQ_3 TQ_4 TQ_2 TQ_1

7

0	0	0	0
0	0	0	1
0	0	1	0
0	0	1	1
0	1	0	0
0	1	0	1
0	1	1	0
0	1	1	1
1	0	0	0
1	0	0	1

~~$Q_3 \bar{Q}_4 \bar{Q}_2 Q_1$~~

$Q_3 \bar{Q}_4$

$\bar{Q}_3 \bar{Q}_4$	$\bar{Q}_3 Q_4$	$Q_3 \bar{Q}_4$	$Q_3 Q_4$
0	0	0	0
0	0	0	0
X	X	X	X
1	1	X	X

$\bar{Q}_3 \bar{Q}_4$

$\bar{Q}_3 \bar{Q}_4$	$\bar{Q}_3 Q_4$	$Q_3 \bar{Q}_4$	$Q_3 Q_4$
0	0	0	0
1	1	1	1
X	X	X	X
0	0		

0	0	1	1
0	0	1	1
X	X	X	X
0	0	X	X

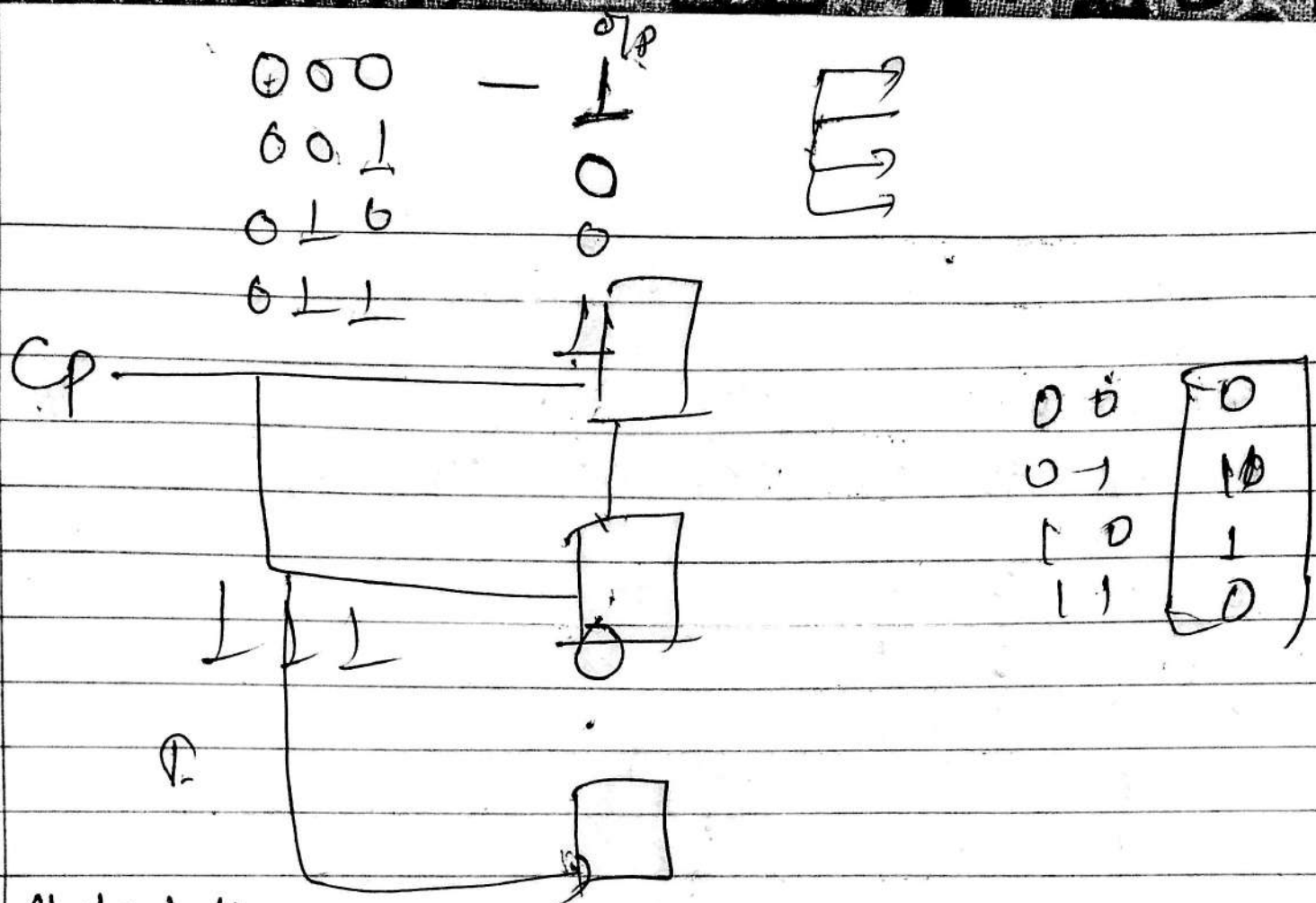
0	1	1	0
0	1	1	0
X	X	X	X
0	1	X	X

0	0	0	0
0	0	0	0
X	X	X	X
0	1	X	X

0	0	0	0
0	0	1	0
X	X	X	X
0	1	X	X

$$TQ_3 = Q_3 Q_1 + Q_1 Q_2 Q_1$$

$Q_3 \bar{Q}_4$



State table

Present state

A B C

0 0 0

0 0 1

0 1 0

0 1 1

1 0 0

1 0 1

1 1 0

1 1 1

Next state

A B C

0 0 1

0 1 1

0 1 0

1 1 0

1 1 1

1 1 0

1 0 0

0 0 0

Q

Q_A

Q_B

Q_C

0

0

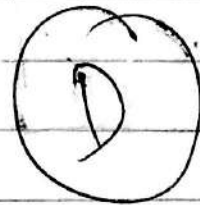
1

Next state

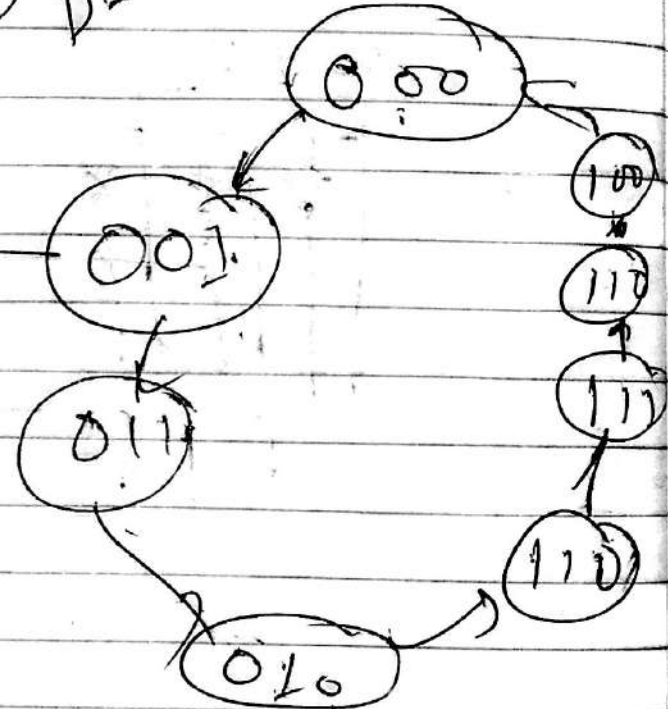
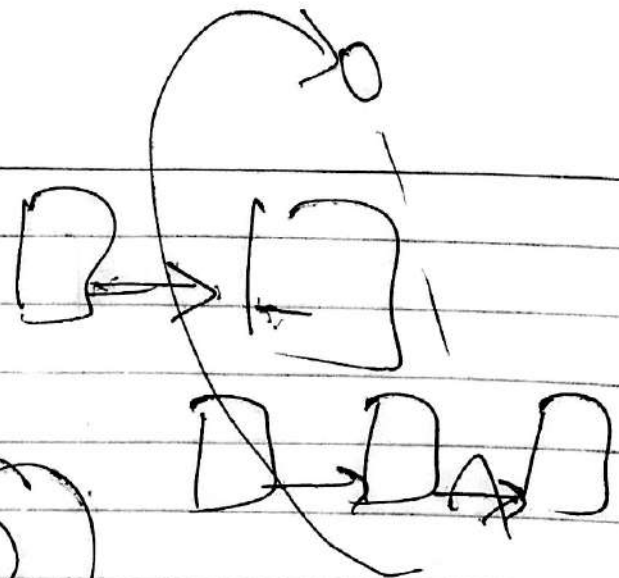
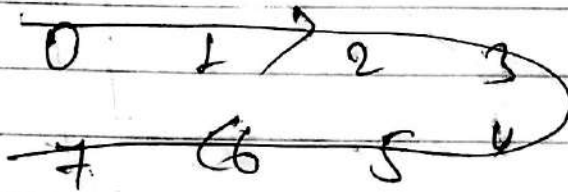
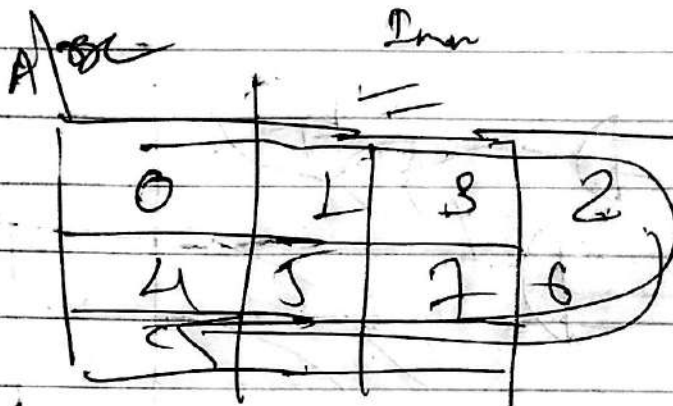
S R

Q_n S R Q_{n+1}

0
0
1
1
0
1

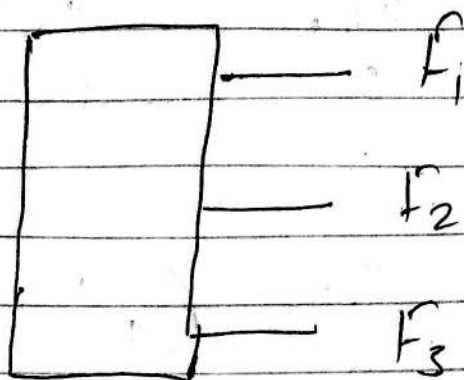


D-



0	0	000
1	1	001
2	3	011
3	2	010
4	6	100
5	7	101
6	5	110
7	4	111

Pre 1 1 1 net 1 1 1



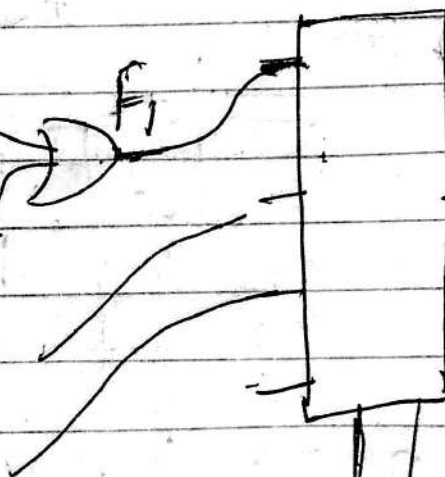
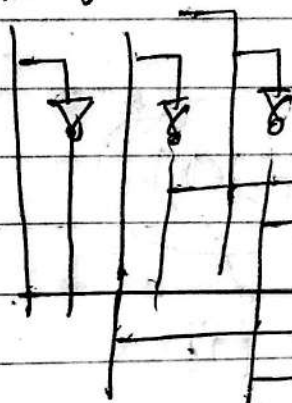
①

2x1

~~4x1~~

3

x y z



F1, F2, F3

2x1

4x1

①

0 0
0 1
1 0
1 1

So

F1

F1, F2, F3

$$\begin{aligned} \textcircled{\bar{A}} &\Rightarrow \bar{A} + 1 = 1 \\ &\quad \bar{A} + 0 = \bar{A} \end{aligned}$$

$$A + B[AC + B\{AC + (B+C)D\}]$$

$$A + B[AC + B\{AC + BD + CD\}]$$

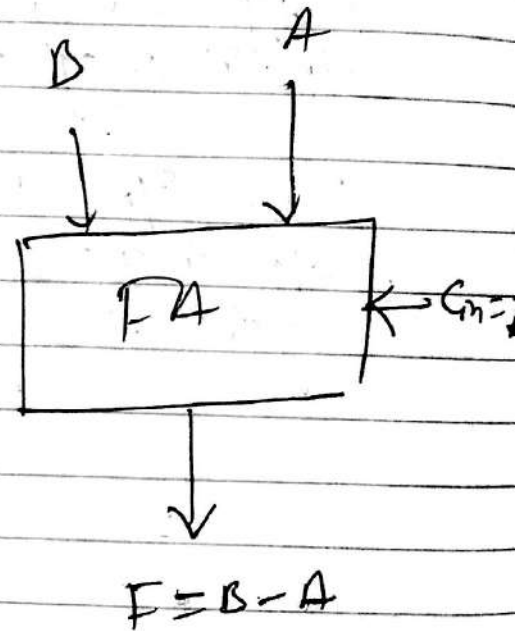
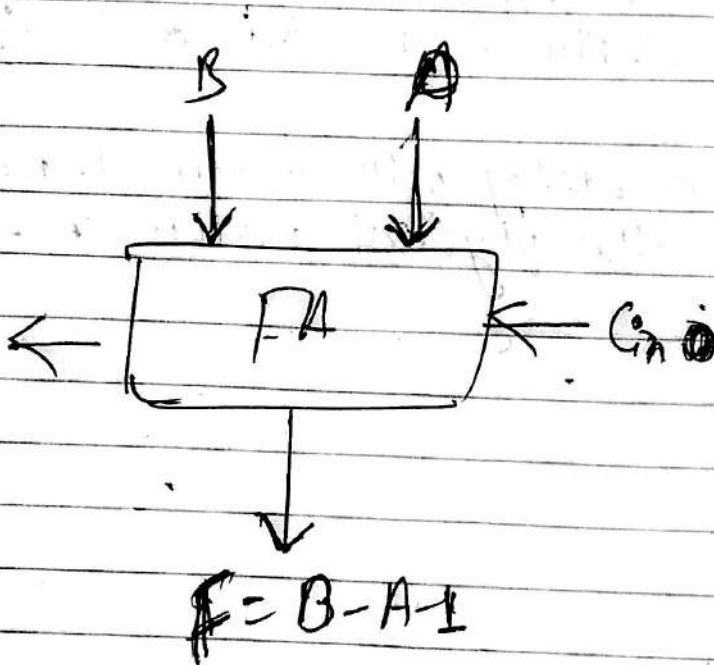
$$A + B[AC + ABC + BD + BC'D]$$

$$A + \underline{ABC} + \underline{ABC} + BD + BC'D$$

$$A + ABC + BD(1+C')$$

$$A + ABC + BD$$

1498



1498

Output Hazard race:

It is the behaviour of an electronics or software system where the o/p is depends on the sequence or

condition may occur in a logical gates where i/p's before the output changes. If a i/p changes a delay will occur.

no. of states

doubled if shift

→ The no. of states can be

named as

register is connected in a shift

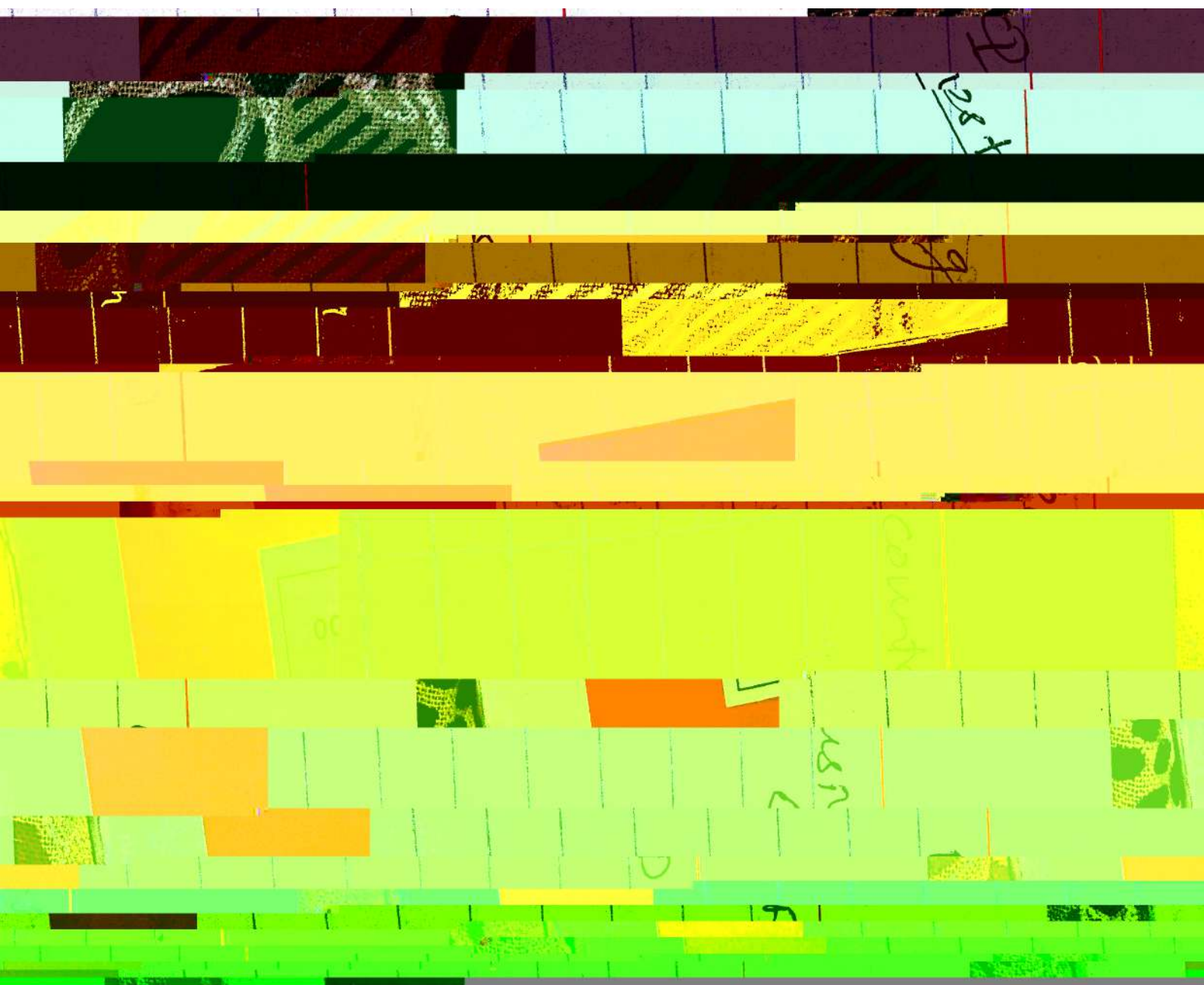
ring counter).

counter is a k-bit shift register.

A switch tail-ring counter is



→ 4-sta



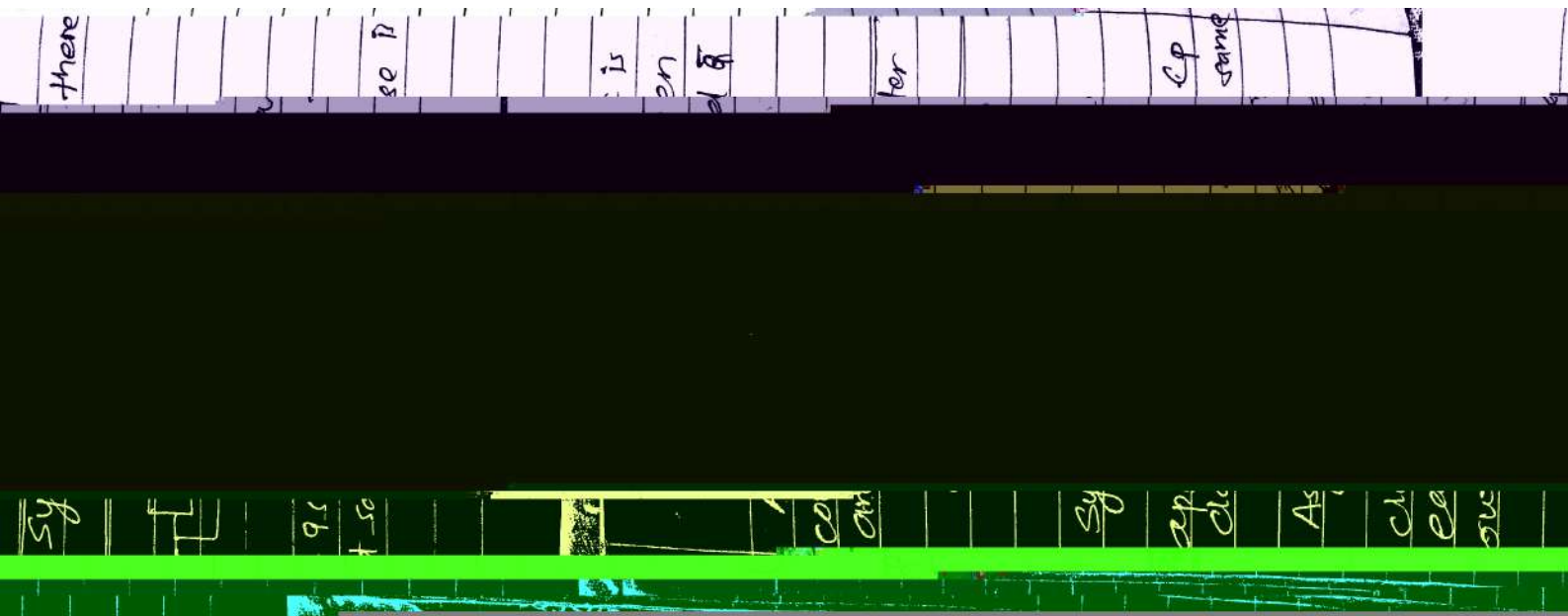


as controlled sequence if
down goes reverse
Similarly if up/down signal is low

for

Question: Design three bit synchronous up/down counter
① T-FF.

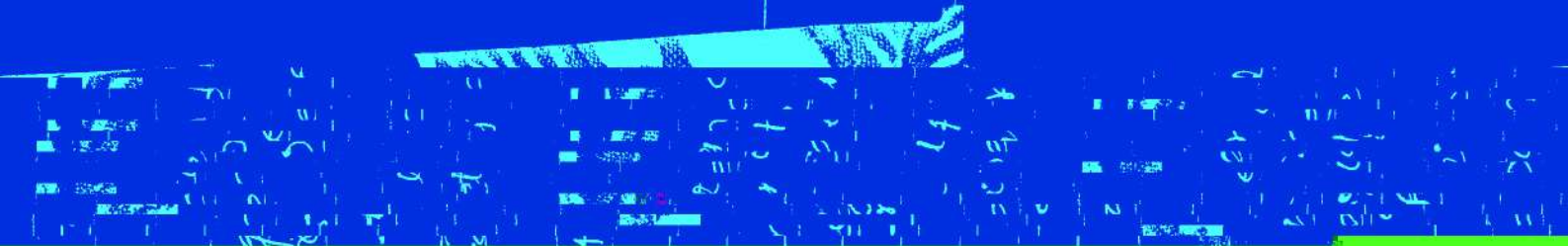
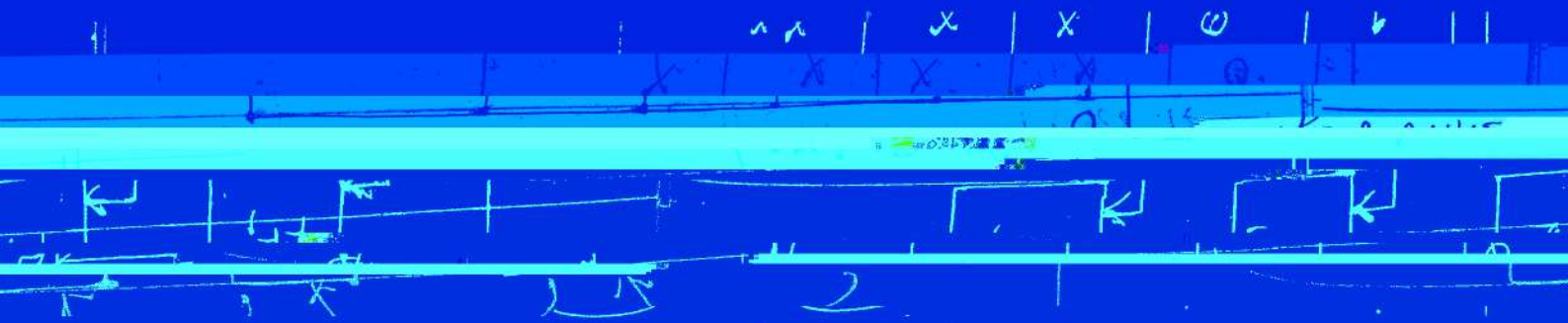
is complex to construct: (b) Estimated with
 at J. T. F.
 slower than system
 counter.

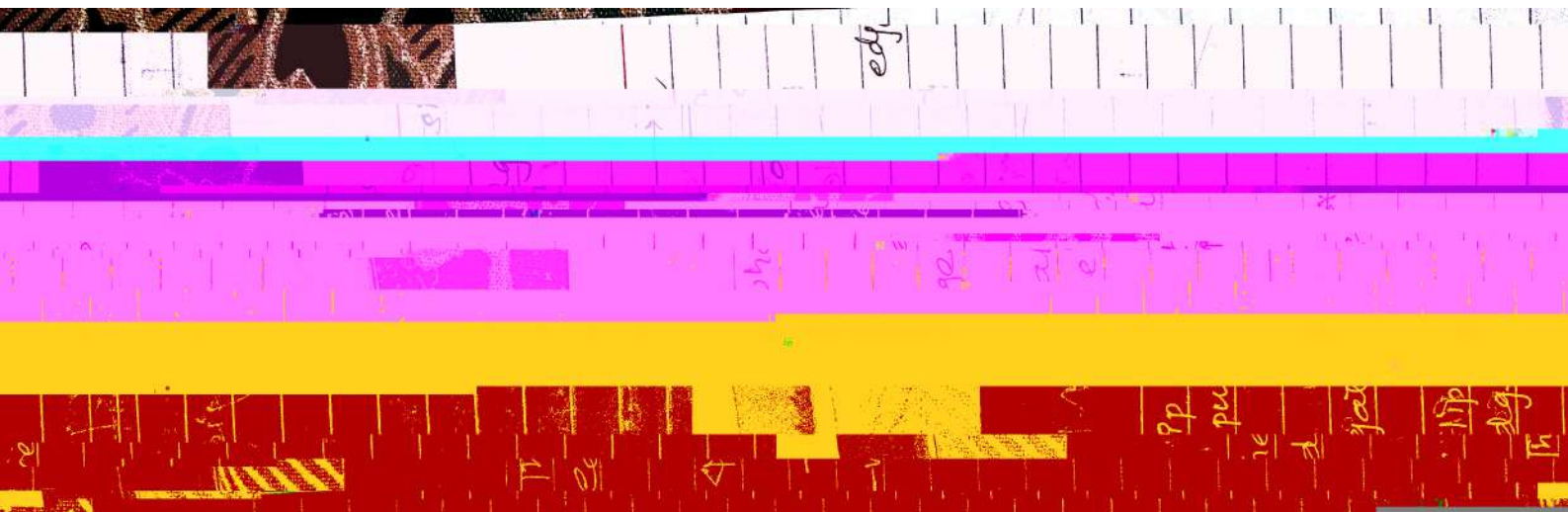


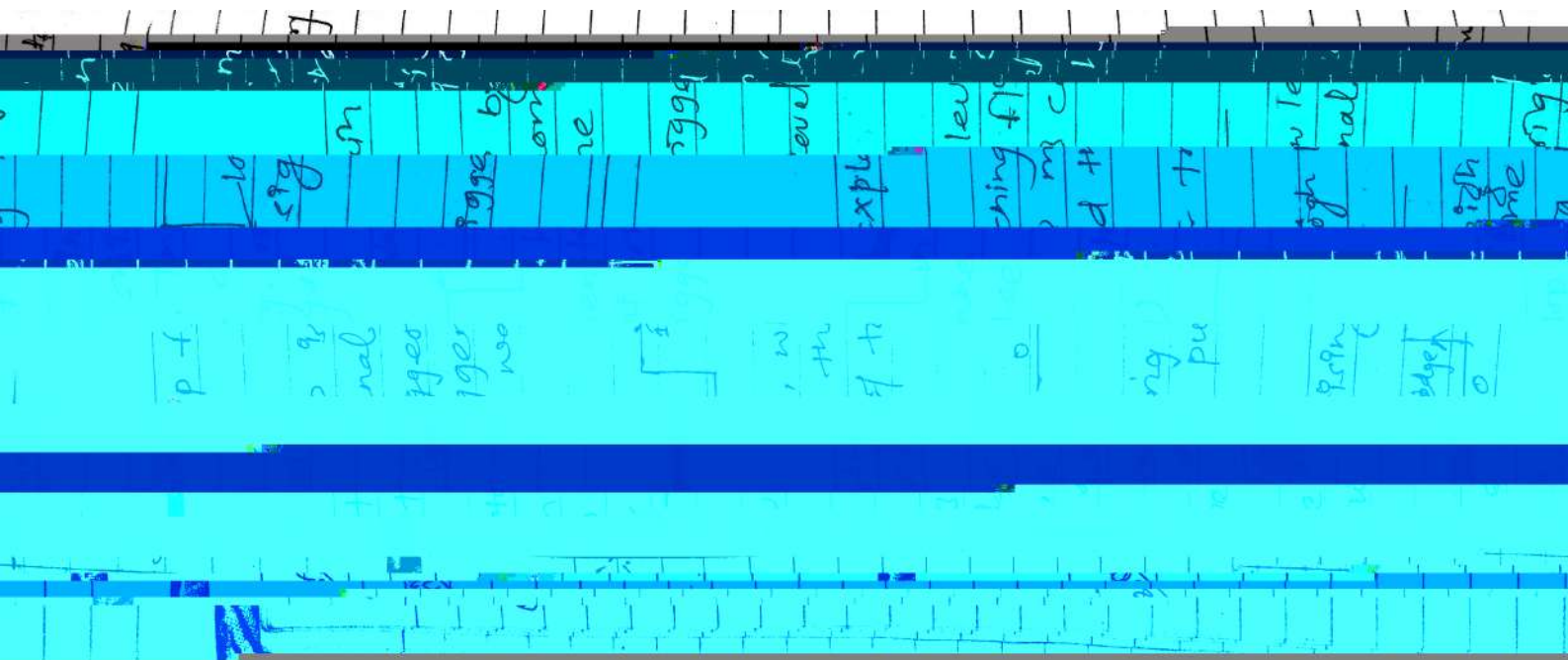


2. PISO Shift Register

Block pulse of each bit appears in 0/1 state







Reduction

minimizing the cost of flip-flops in the sequential circuit is the primary objective of state reduction.

Assignment: To be completed in terms of minimizing the cost of the circuit.

Following are the steps to be followed:

1.

2. 3. 4. 5. 6. 7. 8. 9. 10. 11. 12. 13. 14. 15. 16. 17. 18. 19. 20. 21. 22. 23. 24. 25. 26. 27. 28. 29. 30. 31. 32. 33. 34. 35. 36. 37. 38. 39. 40. 41. 42. 43. 44. 45. 46. 47. 48. 49. 50. 51. 52. 53. 54. 55. 56. 57. 58. 59. 60. 61. 62. 63. 64. 65. 66. 67. 68. 69. 70. 71. 72. 73. 74. 75. 76. 77. 78. 79. 80. 81. 82. 83. 84. 85. 86. 87. 88. 89. 90. 91. 92. 93. 94. 95. 96. 97. 98. 99. 100.

101. 102. 103. 104. 105. 106. 107. 108. 109. 110. 111. 112. 113. 114. 115. 116. 117. 118. 119. 120. 121. 122. 123. 124. 125. 126. 127. 128. 129. 130. 131. 132. 133. 134. 135. 136. 137. 138. 139. 140. 141. 142. 143. 144. 145. 146. 147. 148. 149. 150. 151. 152. 153. 154. 155. 156. 157. 158. 159. 160. 161. 162. 163. 164. 165. 166. 167. 168. 169. 170. 171. 172. 173. 174. 175. 176. 177. 178. 179. 180. 181. 182. 183. 184. 185. 186. 187. 188. 189. 190. 191. 192. 193. 194. 195. 196. 197. 198. 199. 200.

201. 202. 203. 204. 205. 206. 207. 208. 209. 210. 211. 212. 213. 214. 215. 216. 217. 218. 219. 220. 221. 222. 223. 224. 225. 226. 227. 228. 229. 230. 231. 232. 233. 234. 235. 236. 237. 238. 239. 240. 241. 242. 243. 244. 245. 246. 247. 248. 249. 250. 251. 252. 253. 254. 255. 256. 257. 258. 259. 260. 261. 262. 263. 264. 265. 266. 267. 268. 269. 270. 271. 272. 273. 274. 275. 276. 277. 278. 279. 280. 281. 282. 283. 284. 285. 286. 287. 288. 289. 290. 291. 292. 293. 294. 295. 296. 297. 298. 299. 300.

301. 302. 303. 304. 305. 306. 307. 308. 309. 310. 311. 312. 313. 314. 315. 316. 317. 318. 319. 320. 321. 322. 323. 324. 325. 326. 327. 328. 329. 330. 331. 332. 333. 334. 335. 336. 337. 338. 339. 340. 341. 342. 343. 344. 345. 346. 347. 348. 349. 350. 351. 352. 353. 354. 355. 356. 357. 358. 359. 360. 361. 362. 363. 364. 365. 366. 367. 368. 369. 370. 371. 372. 373. 374. 375. 376. 377. 378. 379. 380. 381. 382. 383. 384. 385. 386. 387. 388. 389. 390. 391. 392. 393. 394. 395. 396. 397. 398. 399. 400.

401. 402. 403. 404. 405. 406. 407. 408. 409. 410. 411. 412. 413. 414. 415. 416. 417. 418. 419. 420. 421. 422. 423. 424. 425. 426. 427. 428. 429. 430. 431. 432. 433. 434. 435. 436. 437. 438. 439. 440. 441. 442. 443. 444. 445. 446. 447. 448. 449. 450. 451. 452. 453. 454. 455. 456. 457. 458. 459. 460. 461. 462. 463. 464. 465. 466. 467. 468. 469. 470. 471. 472. 473. 474. 475. 476. 477. 478. 479. 480. 481. 482. 483. 484. 485. 486. 487. 488. 489. 490. 491. 492. 493. 494. 495. 496. 497. 498. 499. 500.

501. 502. 503. 504. 505. 506. 507. 508. 509. 510. 511. 512. 513. 514. 515. 516. 517. 518. 519. 520. 521. 522. 523. 524. 525. 526. 527. 528. 529. 530. 531. 532. 533. 534. 535. 536. 537. 538. 539. 540. 541. 542. 543. 544. 545. 546. 547. 548. 549. 550. 551. 552. 553. 554. 555. 556. 557. 558. 559. 560. 561. 562. 563. 564. 565. 566. 567. 568. 569. 570. 571. 572. 573. 574. 575. 576. 577. 578. 579. 580. 581. 582. 583. 584. 585. 586. 587. 588. 589. 590. 591. 592. 593. 594. 595. 596. 597. 598. 599. 600.

601. 602. 603. 604. 605. 606. 607. 608. 609. 610. 611. 612. 613. 614. 615. 616. 617. 618. 619. 620. 621. 622. 623. 624. 625. 626. 627. 628. 629. 630. 631. 632. 633. 634. 635. 636. 637. 638. 639. 640. 641. 642. 643. 644. 645. 646. 647. 648. 649. 650. 651. 652. 653. 654. 655. 656. 657. 658. 659. 660. 661. 662. 663. 664. 665. 666. 667. 668. 669. 670. 671. 672. 673. 674. 675. 676. 677. 678. 679. 680. 681. 682. 683. 684. 685. 686. 687. 688. 689. 690. 691. 692. 693. 694. 695. 696. 697. 698. 699. 700.

701. 702. 703. 704. 705. 706. 707. 708. 709. 710. 711. 712. 713. 714. 715. 716. 717. 718. 719. 720. 721. 722. 723. 724. 725. 726. 727. 728. 729. 730. 731. 732. 733. 734. 735. 736. 737. 738. 739. 740. 741. 742. 743. 744. 745. 746. 747. 748. 749. 750. 751. 752. 753. 754. 755. 756. 757. 758. 759. 760. 761. 762. 763. 764. 765. 766. 767. 768. 769. 770. 771. 772. 773. 774. 775. 776. 777. 778. 779. 780. 781. 782. 783. 784. 785. 786. 787. 788. 789. 790. 791. 792. 793. 794. 795. 796. 797. 798. 799. 800.

801. 802. 803. 804. 805. 806. 807. 808. 809. 810. 811. 812. 813. 814. 815. 816. 817. 818. 819. 820. 821. 822. 823. 824. 825. 826. 827. 828. 829. 830. 831. 832. 833. 834. 835. 836. 837. 838. 839. 840. 841. 842. 843. 844. 845. 846. 847. 848. 849. 850. 851. 852. 853. 854. 855. 856. 857. 858. 859. 860. 861. 862. 863. 864. 865. 866. 867. 868. 869. 870. 871. 872. 873. 874. 875. 876. 877. 878. 879. 880. 881. 882. 883. 884. 885. 886. 887. 888. 889. 890. 891. 892. 893. 894. 895. 896. 897. 898. 899. 900.

901. 902. 903. 904. 905. 906. 907. 908. 909. 910. 911. 912. 913. 914. 915. 916. 917. 918. 919. 920. 921. 922. 923. 924. 925. 926. 927. 928. 929. 930. 931. 932. 933. 934. 935. 936. 937. 938. 939. 940. 941. 942. 943. 944. 945. 946. 947. 948. 949. 950. 951. 952. 953. 954. 955. 956. 957. 958. 959. 960. 961. 962. 963. 964. 965. 966. 967. 968. 969. 970. 971. 972. 973. 974. 975. 976. 977. 978. 979. 980. 981. 982. 983. 984. 985. 986. 987. 988. 989. 990. 991. 992. 993. 994. 995. 996. 997. 998. 999. 1000.

1001. 1002. 1003. 1004. 1005. 1006. 1007. 1008. 1009. 1010. 1011. 1012. 1013. 1014. 1015. 1016. 1017. 1018. 1019. 1020. 1021. 1022. 1023. 1024. 1025. 1026. 1027. 1028. 1029. 1030. 1031. 1032. 1033. 1034. 1035. 1036. 1037. 1038. 1039. 1040. 1041. 1042. 1043. 1044. 1045. 1046. 1047. 1048. 1049. 1050. 1051. 1052. 1053. 1054. 1055. 1056. 1057. 1058. 1059. 1060. 1061. 1062. 1063. 1064. 1065. 1066. 1067. 1068. 1069. 1070. 1071. 1072. 1073. 1074. 1075. 1076. 1077. 1078. 1079. 1080. 1081. 1082. 1083. 1084. 1085. 1086. 1087. 1088. 1089. 1090. 1091. 1092. 1093. 1094. 1095. 1096. 1097. 1098. 1099. 1100.

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1201. 1202. 1203. 1204. 1205. 1206. 1207. 1208. 1209. 1210. 1211. 1212. 1213. 1214. 1215. 1216. 1217. 1218. 1219. 1220. 1221. 1222. 1223. 1224. 1225. 1226. 1227. 1228. 1229. 1230. 1231. 1232. 1233. 1234. 1235. 1236. 1237. 1238. 1239. 1240. 1241. 1242. 1243. 1244. 1245. 1246. 1247. 1248. 1249. 1250. 1251. 1252. 1253. 1254. 1255. 1256. 1257. 1258. 1259. 1260. 1261. 1262. 1263. 1264. 1265. 1266. 1267. 1268. 1269. 1270. 1271. 1272. 1273. 1274. 1275. 1276. 1277. 1278. 1279. 1280. 1281. 1282. 1283. 1284. 1285. 1286. 1287. 1288. 1289. 1290. 1291. 1292. 1293. 1294. 1295. 1296. 1297. 1298. 1299. 1300.

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1501. 1502. 1503. 1504. 1505. 1506. 1507. 1508. 1509. 1510. 1511. 1512. 1513. 1514. 1515. 1516. 1517. 1518. 1519. 1520. 1521. 1522. 1523. 1524. 1525. 1526. 1527. 1528. 1529. 1530. 1531. 1532. 1533. 1534. 1535. 1536. 1537. 1538. 1539. 1540. 1541. 1542. 1543. 1544. 1545. 1546. 1547. 1548. 1549. 1550. 1551. 1552. 1553. 1554. 1555. 1556. 1557. 1558. 1559. 1560. 1561. 1562. 1563. 1564. 1565. 1566. 1567. 1568. 1569. 1570. 1571. 1572. 1573. 1574. 1575. 1576. 1577. 1578. 1579. 1580. 1581. 1582. 1583. 1584. 1585. 1586. 1587. 1588. 1589. 1590. 1591. 1592. 1593. 1594. 1595. 1596. 1597. 1598. 1599. 1600.

1601. 1602. 1603. 1604. 1605. 1606. 1607. 1608. 1609. 1610. 1611. 1612. 1613. 1614. 1615. 1616. 1617. 1618. 1619. 1620. 1621. 1622. 1623. 1624. 1625. 1626. 1627. 1628. 1629. 1630. 1631. 1632. 1633. 1634. 1635. 1636. 1637. 1638. 1639. 1640. 1641. 1642. 1643. 1644. 1645. 1646. 1647. 1648. 1649. 1650. 1651. 1652. 1653. 1654. 1655. 1656. 1657. 1658. 1659. 1660. 1661. 1662. 1663. 1664. 1665. 1666. 1667. 1668. 1669. 1670. 1671. 1672. 1673. 1674. 1675. 1676. 1677. 1678. 1679. 1680. 1681. 1682. 1683. 1684. 1685. 1686. 1687. 1688. 1689. 1690. 1691. 1692. 1693. 1694. 1695. 1696. 1697. 1698. 1699. 1700.

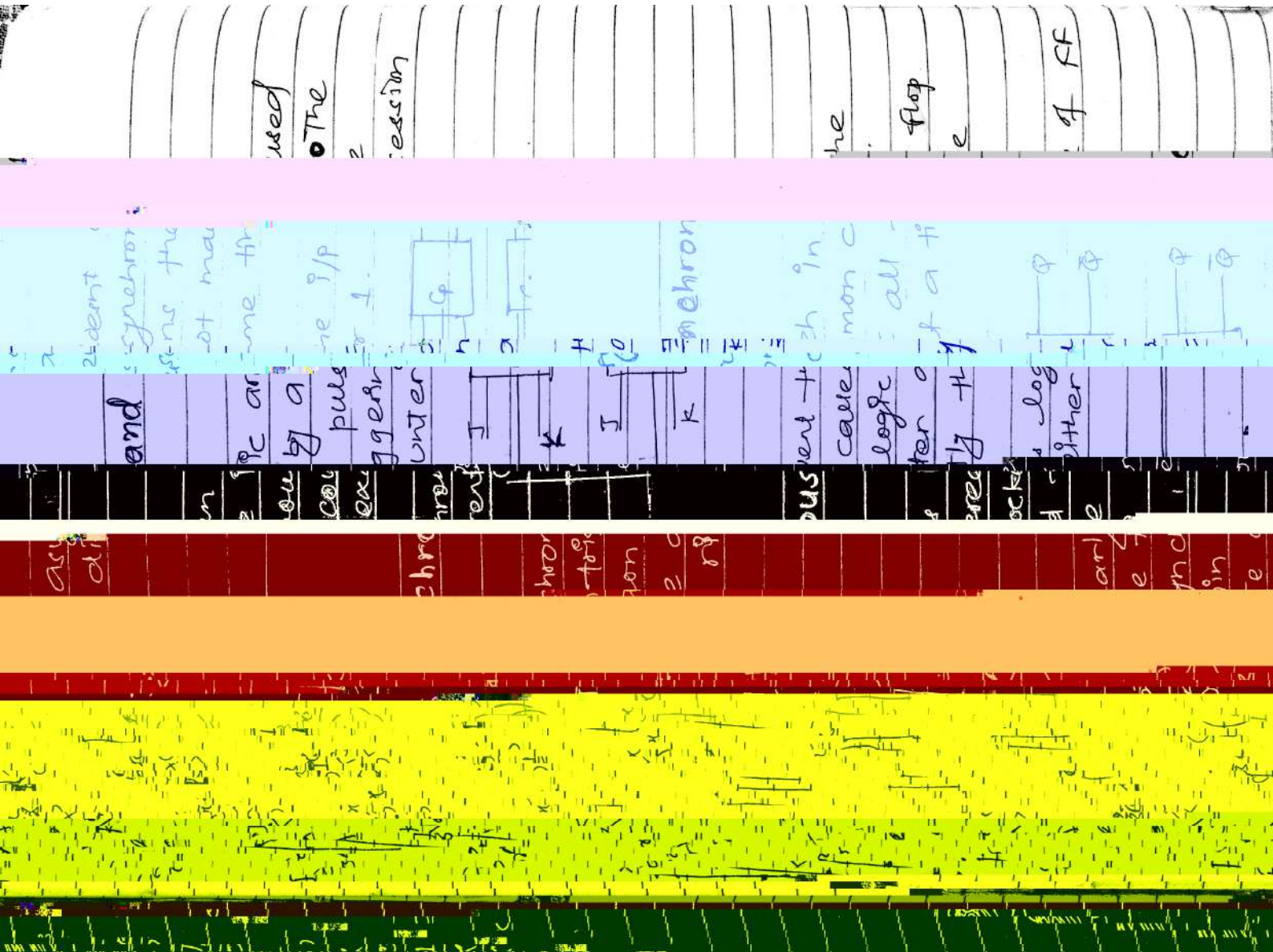
1701. 1702. 1703. 1704. 1705. 1706. 1707. 1708. 1709. 1710. 1711. 1712. 1713. 1714. 1715. 1716. 1717. 1718. 1719. 1720. 1721. 1722. 1723. 1724. 1725. 1726. 1727. 1728. 1729. 1730. 1731. 1732. 1733. 1734. 1735. 1736. 1737. 1738. 1739. 1740. 1741. 1742. 1743. 1744. 1745. 1746. 1747. 1748. 1749. 1750. 1751. 1752. 1753. 1754. 1755. 1756. 1757. 1758. 1759. 1760. 1761. 1762. 1763. 1764. 1765. 1766. 1767. 1768. 1769. 1770. 1771. 1772. 1773. 1774. 1775. 1776. 1777. 1778. 1779. 1780. 1781. 1782. 1783. 1784. 1785. 1786. 1787. 1788. 1789. 1790. 1791. 1792. 1793. 1794. 1795. 1796. 1797. 1798. 1799. 1800.

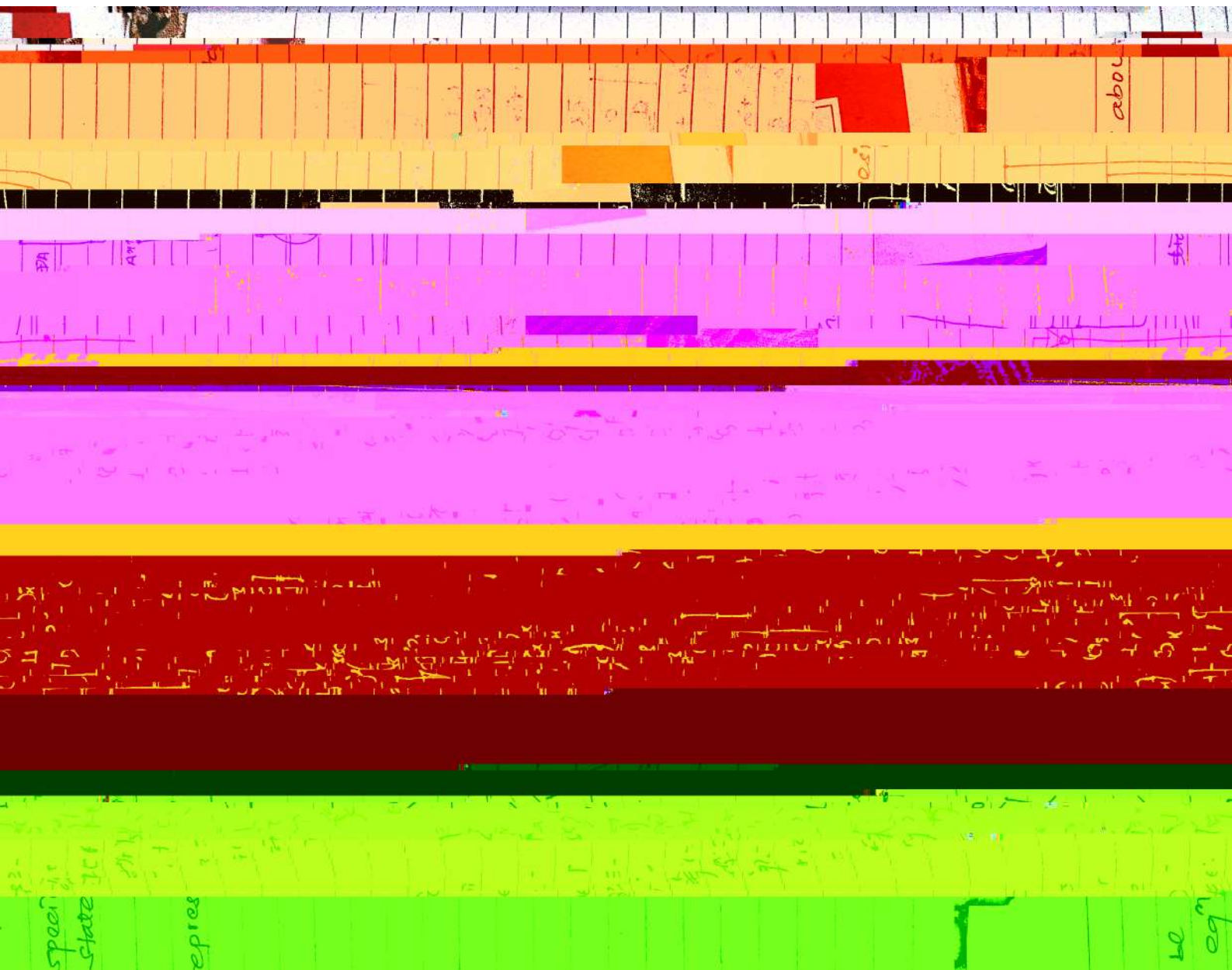
1801. 1802. 1803. 1804. 1805. 1806. 1807. 1808. 1809. 1810. 1811. 1812. 1813. 1814. 1815. 1816. 1817. 1818. 1819. 1820. 1821. 1822. 1823. 1824. 1825. 1826. 1827. 1828. 1829. 1830. 1831. 1832. 1833. 1834. 1835. 1836. 1837. 1838. 1839. 1840. 1841. 1842. 1843. 1844. 1845. 1846. 1847. 1848. 1849. 1850. 1851. 1852. 1853. 1854. 1855. 1856. 1857. 1858. 1859. 1860. 1861. 1862. 1863. 1864. 1865. 1866. 1867. 1868. 1869. 1870. 1871. 1872. 1873. 1874. 1875. 1876. 1877. 1878. 1879. 1880. 1881. 1882. 1883. 1884. 1885. 1886. 1887. 1888. 1889. 1890. 1891. 1892. 1893. 1894. 1895. 1896. 1897. 1898. 1899. 1900.

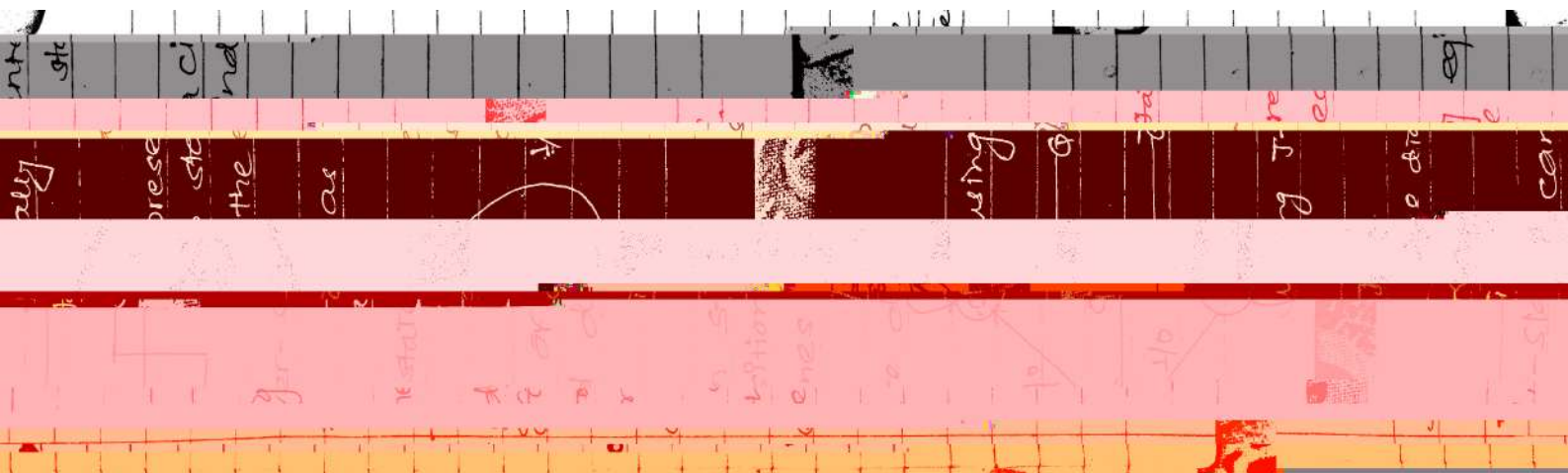
1901. 1902. 1903. 1904. 1905. 1906. 1907. 1908. 1909. 1910. 1911. 1912. 1913. 1914. 1915. 1916. 1917. 1918. 1919. 1920. 1921. 1922. 1923. 1924. 1925. 1926. 1927. 1928. 1929. 1930. 1931. 1932. 1933. 1934. 1935. 1936. 1937. 1938. 1939. 1940. 1941. 1942. 1943. 1944. 1945. 1946. 1947. 1948. 1949. 1950. 1951. 1952. 1953. 1954. 1955. 1956. 1957. 1958. 1959. 1960. 1961. 1962. 1963. 1964. 1965. 1966. 1967. 1968. 1969. 1970. 1971. 1972. 1973. 1974. 1975. 1976. 1977. 1978. 1979. 1980. 1981. 1982. 1983. 1984. 1985. 1986. 1987. 1988. 1989. 1990. 1991. 1992. 1993. 1994. 1995. 1996. 1997. 1998. 1999. 2000.

2001. 2002. 2003. 2004. 2005. 2006. 2007. 2008. 2009. 2010. 2011. 2012. 2013. 2014. 2015. 2016. 2017. 2018. 2019. 2020. 2021. 2022. 2023. 2024. 2025. 2026. 2027. 2028. 2029. 2030. 2031. 2032. 2033. 2034. 2035. 2036. 2037. 2038. 2039. 2040. 2041. 2042. 2043. 2044. 2045. 2046. 2047. 2048. 2049. 2050. 2051. 2052. 2053. 2054. 2055. 2056. 2057. 2058. 2059. 2060. 2061. 2062. 2063. 2064. 2065. 2066. 2067. 2068. 2069. 2070. 2071. 2072. 2073. 2074. 2075. 2076. 2077. 2078. 2079. 2080. 2081. 2082. 2083. 2084. 2085. 2086. 2087. 2088. 2089. 2090. 2091. 2092. 2093. 2094. 2095. 2096. 2097. 2098. 2099. 2100.

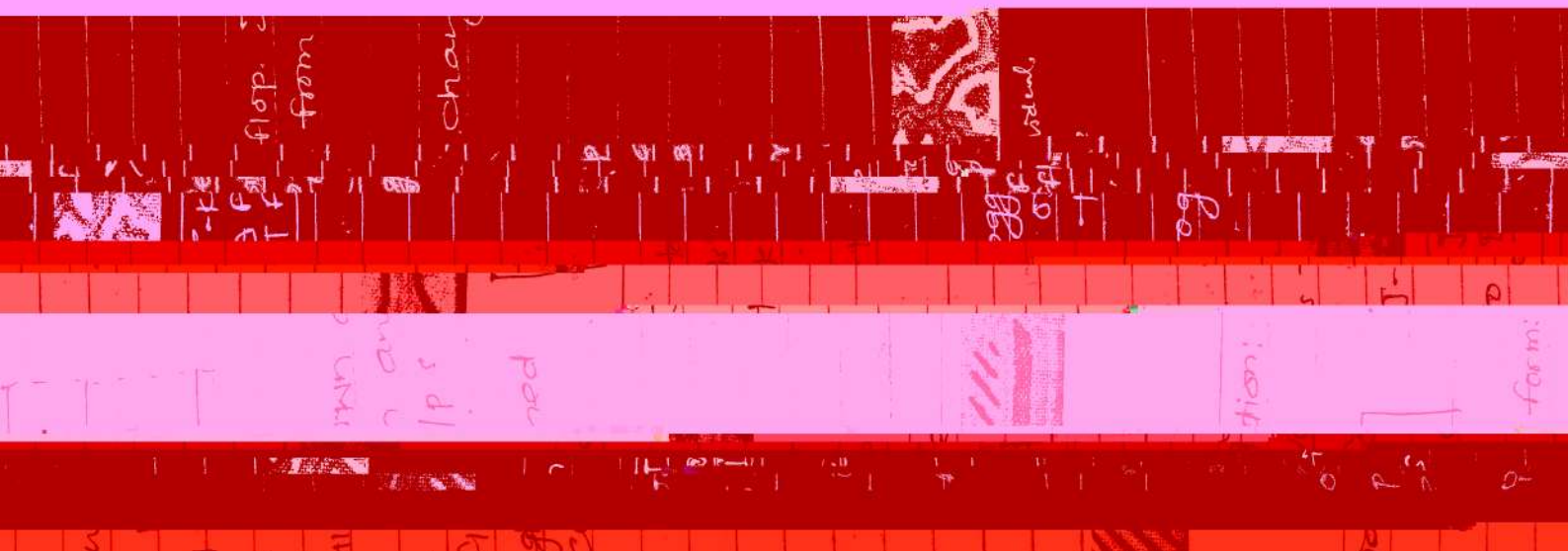
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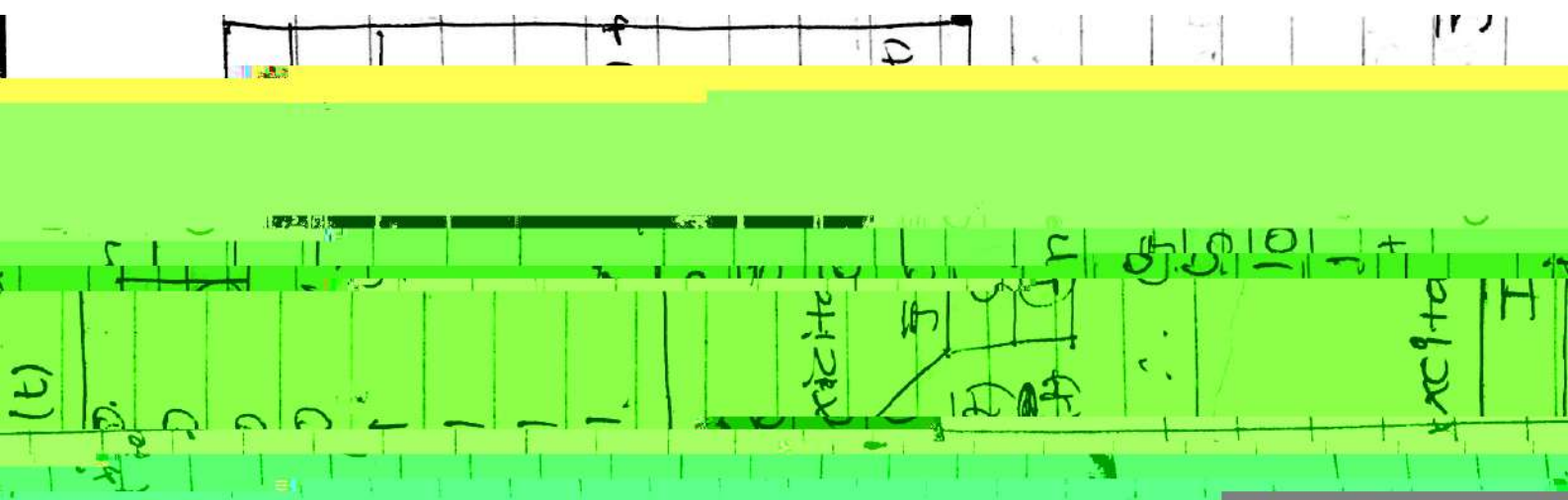


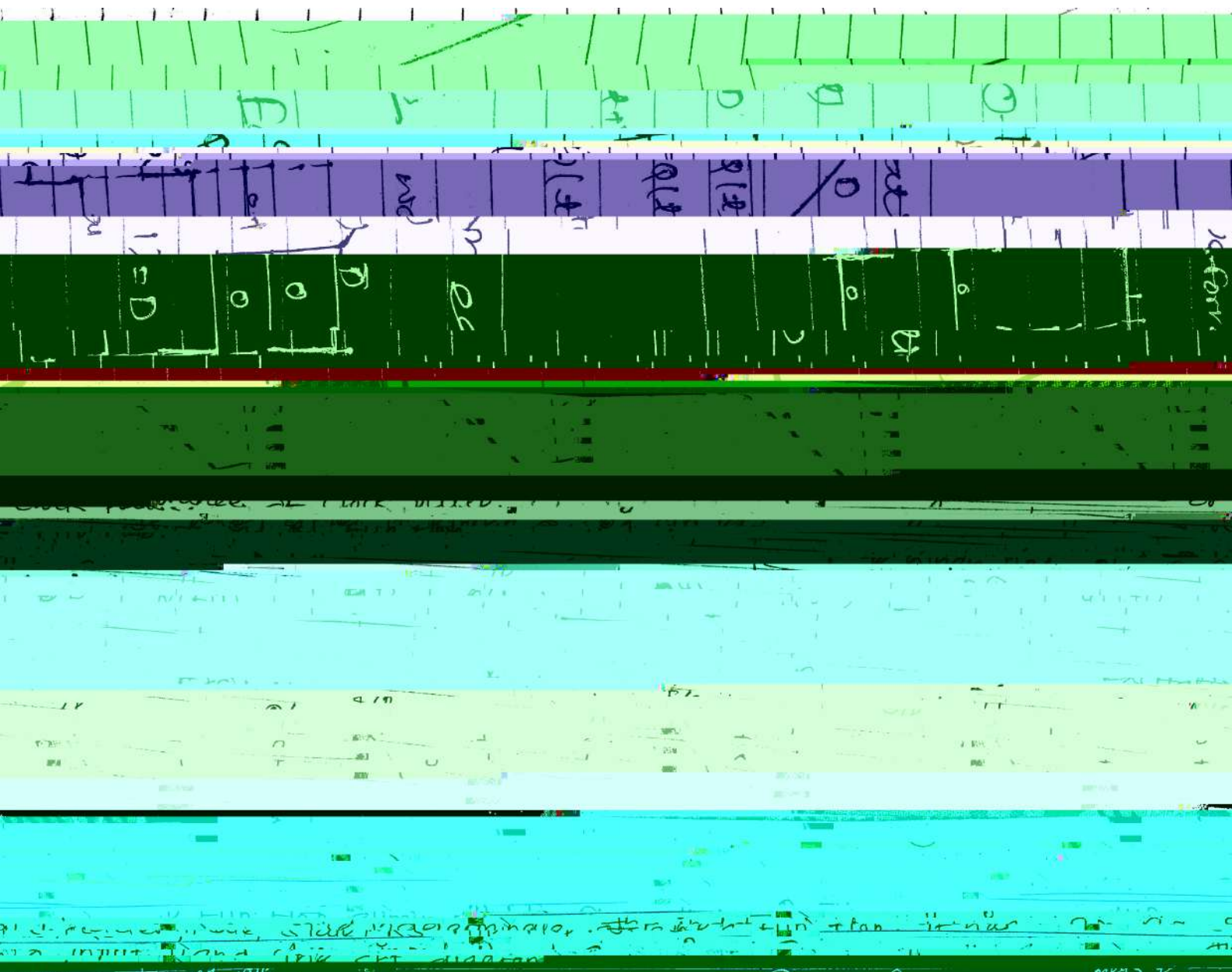




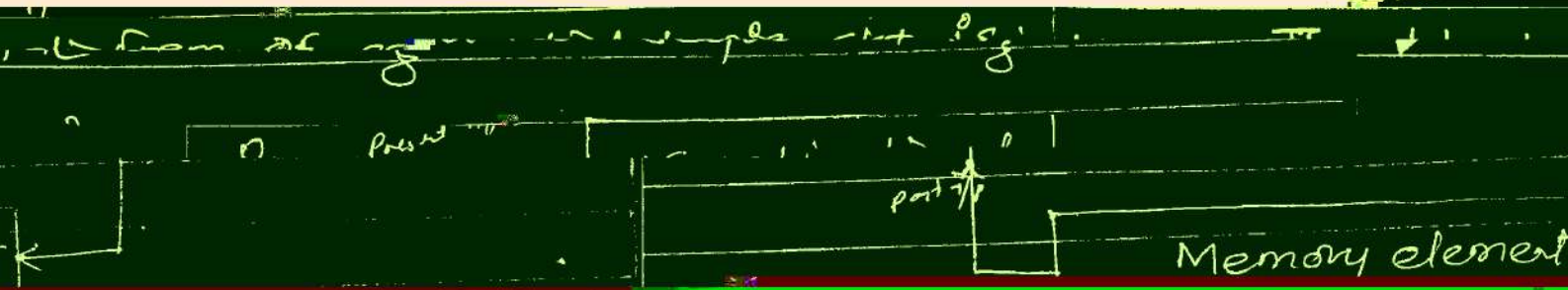
FF, 0
s sla
no ff
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and
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master







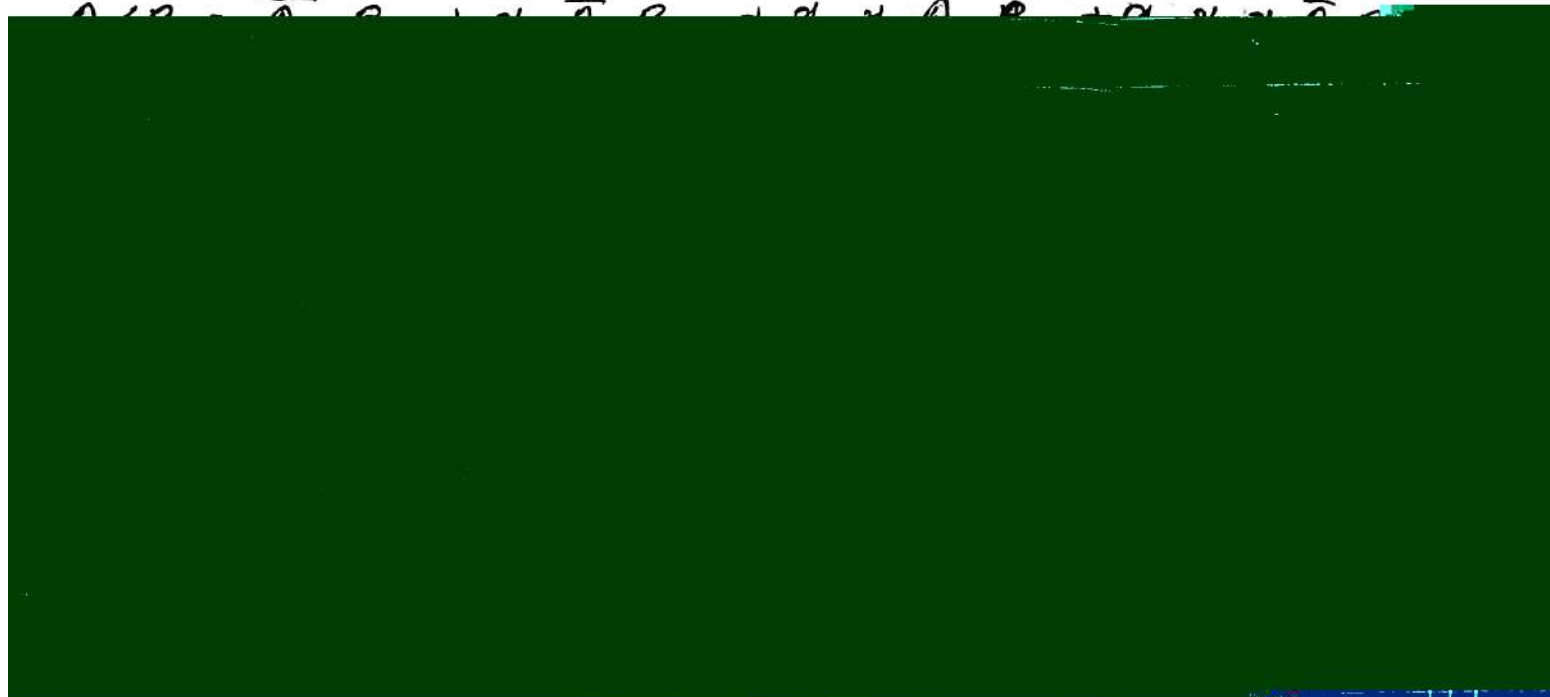
Q. A sequential logic circuit consist of a logic ckt to



which stores the memory at any time, gives

A < B.

for



inter

Is a combinational circuit that compares
A and B and determines their relative

→ It
two ne
mag. :-

$$= B_2, A_1 = B_1$$

$$A_4 = B_4, A_3 = B_3, A_2 =$$

$$x_i = A_i B_i + A_i \bar{B}_i \quad (\text{when } i=1, 2, 3, 4)$$

are equal.

$$\text{ie, } A=B = x_4 x_3 x_2 x_1$$

→ The variable $A=B=1$ if all pairs of digits of two numbers are equal.

Handwritten notes on a lined background, including mathematical expressions and diagrams related to the 4-bit comparator circuit.

Handwritten notes on a lined background, including mathematical expressions and diagrams related to the 4-bit comparator circuit.

Handwritten notes on a lined background, including mathematical expressions and diagrams related to the 4-bit comparator circuit.

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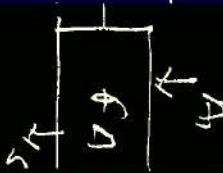
A
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5.

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9

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carrom

$$BC + ABC + ABC$$

$$= \sum(7, 5, 3)$$

Now the k-map is

| | | BC | | | |
|---|---|----|---|---|---|
| A | 0 | 0 | 1 | 1 | 0 |
| | 1 | 0 | 1 | 1 | 0 |

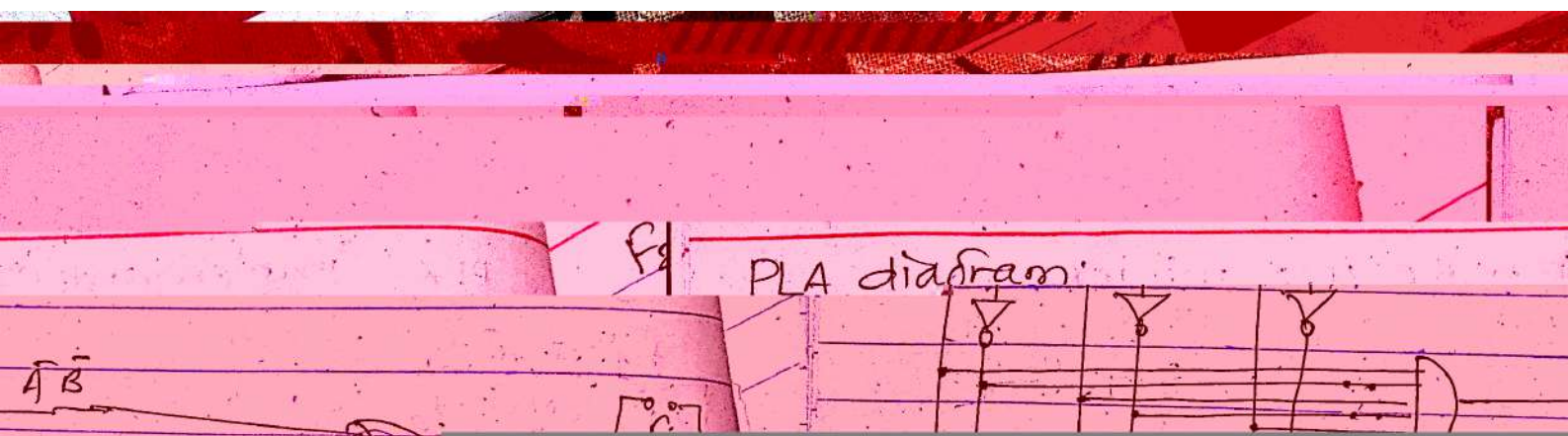
| | | BC | | | |
|---|---|----|---|---|---|
| A | 0 | 0 | 0 | 0 | 0 |
| | 1 | 1 | 1 | 1 | 1 |

$$F_1 + F_2$$

to less no of product term

$$F_1 = AB + AC$$

$$F_2 = AC + AB$$



for PLA implementation we use minimum no. of product term so for that,

F_1 and F_2 : The no. of product terms = 6

F_1 and $\bar{F}_2$: " " " " = 6

$\bar{F}_1$ and F_2 : " " " " = 6

F_1 and $\bar{F}_2$:

from above representation, taking $\bar{F}_1$ complement, and F_2 due to less no. of product term is F_1

$$F_1 = \bar{A}\bar{B} + \bar{B}\bar{C} + \bar{A}C$$

$$F_2 = \bar{B}\bar{C} + \bar{A}C + ABC$$

Now PLA table

| Product terms | A | B | C | F_1 | F_2 |
|------------------|---|---|---|-------|-------|
| $\bar{A}\bar{B}$ | 0 | 0 | 1 | 1 | 0 |
| $\bar{A}\bar{C}$ | 0 | 1 | 0 | 1 | 1 |
| $\bar{B}\bar{C}$ | 1 | 0 | 0 | 1 | 1 |
| $A\bar{B}C$ | 1 | 1 | 1 | 0 | 1 |

Question:

A combinational ckt is defined by the function

$$F_1(A, B, C) = \Sigma(3, 5, 6, 7)$$

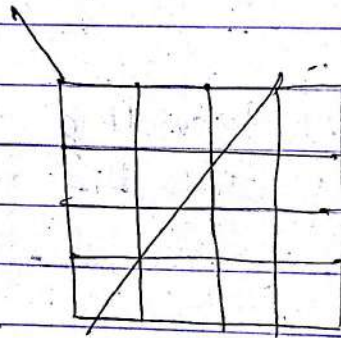
$$F_2(A, B, C) = \Sigma(0, 2, 4, 7)$$

Implement by using PLA.

Solution:

Making boolean function of above canonical form by using k-map

ie,



| | | | | | |
|---|-----------|------------------|------------|------------|------|
| | | BC | | | |
| | | $\bar{B}\bar{C}$ | $\bar{B}C$ | $B\bar{C}$ | BC |
| A | $\bar{A}$ | 0 | 0 | 1 | 0 |
| | A | 0 | 1 | 1 | 1 |

$\bar{F}_1 = \text{group of 0.}$

$F_1 = 1111$

$$F_1 = \bar{A}BC + A\bar{B}C + AB\bar{C} + ABC$$

$$\bar{F}_1 = \bar{A}\bar{B} + \bar{B}\bar{C} + \bar{A}\bar{C}$$

| | | | | | |
|---|-----------|------------------|------------|------------|------|
| | | BC | | | |
| | | $\bar{B}\bar{C}$ | $\bar{B}C$ | $B\bar{C}$ | BC |
| A | $\bar{A}$ | 1 | 0 | 0 | 1 |
| | A | 1 | 0 | 1 | 0 |

$$F_2 = \bar{B}\bar{C} + \bar{A}\bar{C} + \bar{A}BC$$

$$\bar{F}_2 = \bar{B}C + \bar{A}C + A\bar{B}\bar{C}$$

PLA

(Programmable Logic Array)

↳ It is a combinational ckt which may have don't care condition.

↳ It is useful when the no. of don't care condition is excessive.

↳ It doesn't provide full decoding of the variables - and doesn't contains all the m terms as it in the ~~an~~ ROM.

↳ In PLA the decoder is represented by group of AND gate and OR gate.

↳ The block diagram of PLA is

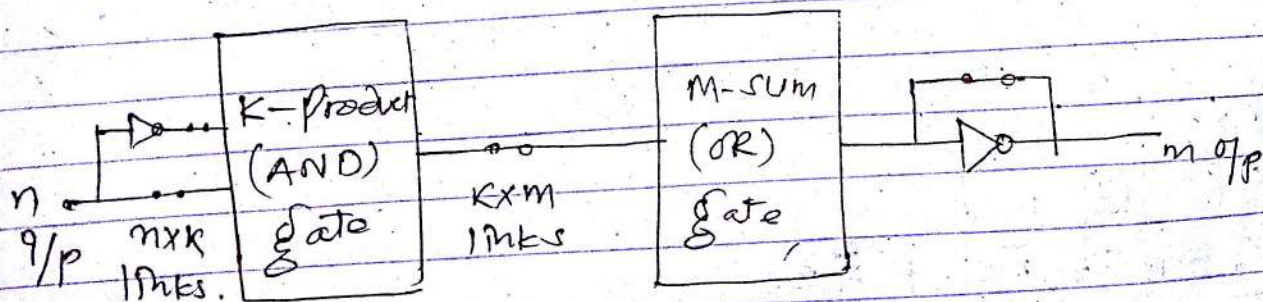
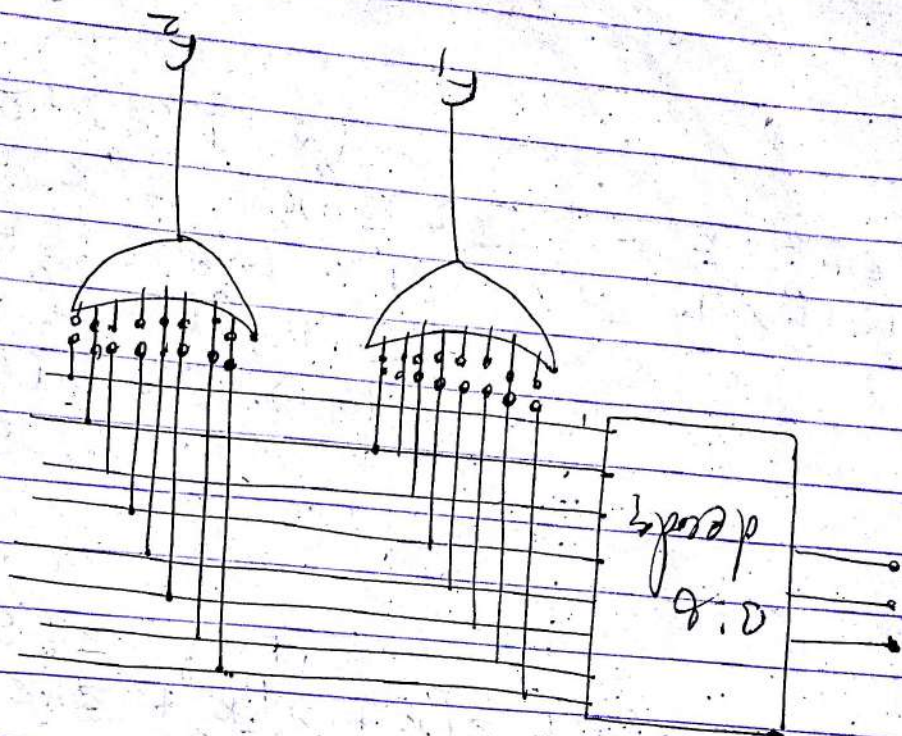


Fig: Block diagram of PLA.



for RNN we

| x | y | z |
|-------|-------|-----|
| 0 | 0 | 0 |
| 0 | 0 | 0 |
| 0 | 0 | 0 |
| 1 | 1 | 0 |
| 0 | 0 | 1 |
| 1 | 0 | 1 |
| 0 | 1 | 1 |
| 1 | 1 | 1 |
| f_1 | f_2 | |
| 0 | 0 | |
| 1 | 0 | |
| 0 | 1 | |
| 1 | 1 | |
| 0 | 0 | |
| 1 | 0 | |
| 0 | 1 | |
| 1 | 1 | |

we for ROM which have three inputs and upto 8 lines are used for that we have 2:8 decoder and we have two outputs, so used two OR gates. Then the ckt diagram or full adder using ROM is shown above.

Question 3 Programmable ROM

Construct PROM for: $f_1 = x + \bar{y}z$
 $f_2 = \bar{x} + y\bar{z}$

The standard SOP is

$$\begin{aligned} f_1 &= x(y + \bar{y}) + \bar{y}z(x + \bar{x}) \\ &= xy + x\bar{y} + x\bar{y}z + \bar{x}\bar{y}z \\ &= xy(z + \bar{z}) + x\bar{y}(z + \bar{z}) + x\bar{y}z + \bar{x}\bar{y}z \\ &= xyz + xy\bar{z} + x\bar{y}z + x\bar{y}\bar{z} + \bar{x}\bar{y}z + \bar{x}\bar{y}\bar{z} \\ &= \Sigma(7, 6, 5, 4, 1) \end{aligned}$$

Again for f_2

$$\begin{aligned} f_2 &= \bar{x} + y\bar{z} \\ &= \bar{x}(y + \bar{y}) + y\bar{z}(x + \bar{x}) \\ &= \bar{x}y + \bar{x}\bar{y} + x\bar{y}\bar{z} + \bar{x}\bar{y}\bar{z} \\ &= \bar{x}y(z + \bar{z}) + \bar{x}\bar{y}(z + \bar{z}) + x\bar{y}\bar{z} + \bar{x}\bar{y}\bar{z} \\ &= \bar{x}yz + \bar{x}y\bar{z} + \bar{x}\bar{y}z + \bar{x}\bar{y}\bar{z} + x\bar{y}\bar{z} + \bar{x}\bar{y}\bar{z} \\ &= \Sigma(3, 2, 6, 1, 0) \end{aligned}$$



ROM: Decoder + OR

Question 2:

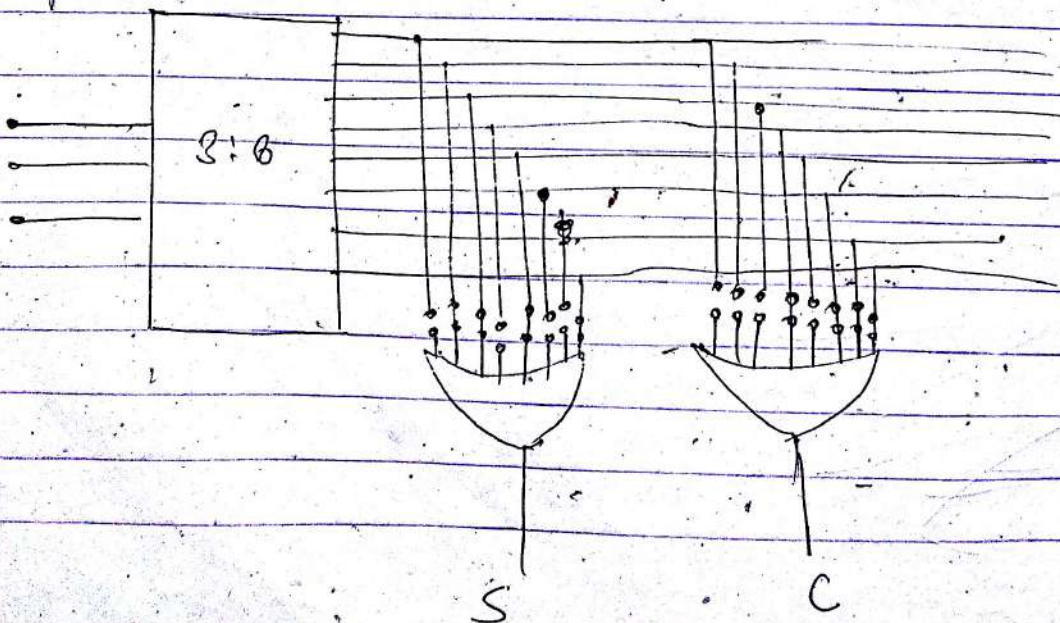
Implement full adder using ROM.

Solution:

The truth table for full adder

| I/p | | | O/p | |
|-----|---|---|-----|-------|
| A | B | C | Sum | Carry |
| 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 | 0 |
| 0 | 1 | 0 | 1 | 0 |
| 0 | 1 | 1 | 0 | 1 |
| 1 | 0 | 0 | 1 | 0 |
| 1 | 0 | 1 | 0 | 1 |
| 1 | 1 | 0 | 0 | 1 |
| 1 | 1 | 1 | 1 | 1 |

Q



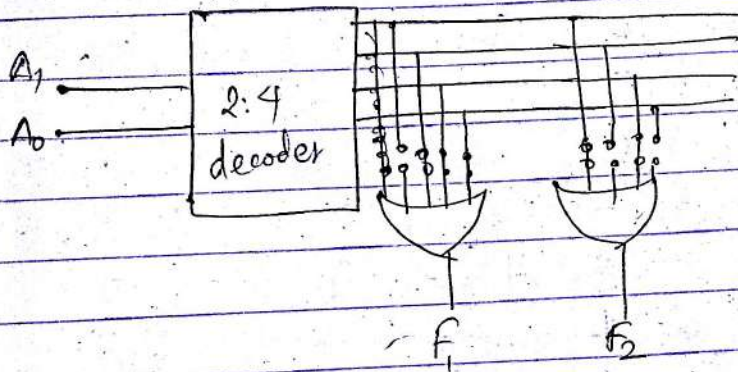
Question:

Implement given boolean function using ROM. (using PROM, EPROM, EEPROM).

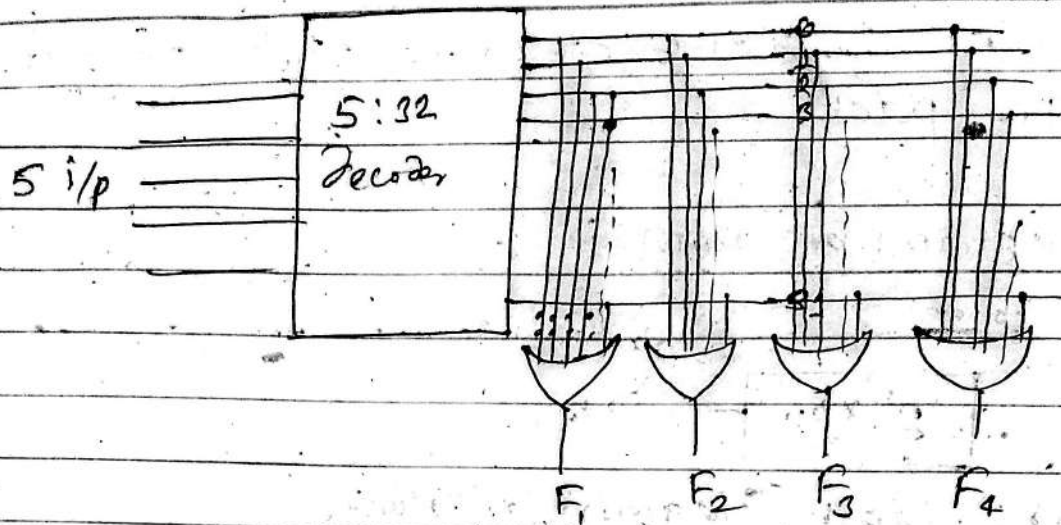
$$F_1(A_1, A_0) = \Sigma(1, 2, 3)$$

$$F_2(A_1, A_0) = \Sigma(0, 2)$$

Ans: 1

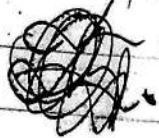


| I/p | | O/p | |
|-------|-------|-------|-------|
| A_1 | A_0 | F_1 | F_2 |
| 0 | 0 | 0 | 1 |
| 0 | 1 | 1 | 0 |
| 1 | 0 | 1 | 1 |
| 1 | 1 | 1 | 0 |



The internal logic construction of 32×4 ROM which have 5 input variables that are decoded as $2^5 = 32$ lines.

The 32 outputs of decoder are connected through switch/fuse to each OR gate. Only 4 of the switches are shown in fig but actually each OR gate has 32 inputs i.e. total switch in 32×4 ROM is 128. which represents the size of the ROM.



→ ROM includes both the decoder and OR gate within a single IC pack.

→ Block Diagram of ROM is

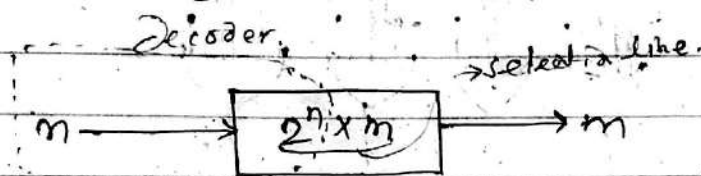


fig: block diagram of ROM.

→ A ROM consists of n i/p lines and m o/p lines. Each bit combination of i/p variable is called address. Each bit combination comes out from the output lines called word.

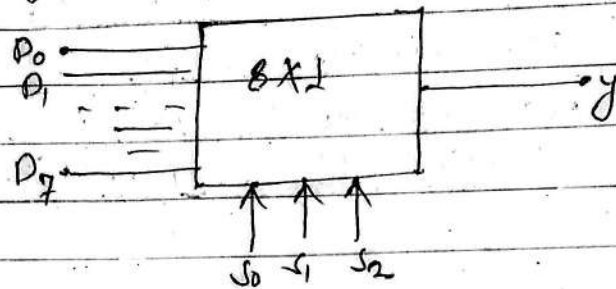
→ The possible address for n input is 2^n .

→ example:

Consider a 32×4 ROM which consists of 32 words of 4 bit, it has m o/p lines and 32 distinct words stored in each unit.

is 5-selection lines so we use 5:32 decoder & 4 OR gate for 4 o/p.

The block diagram is



ROM $n \rightarrow 2^n \times m \rightarrow m$

↳ A Read Only Memory is a semiconductor memory used to store the information permanently. A fixed set of binary information is stored.

↳ It performs only read operation and ^{it} "doesn't" have write operation.

↳ The binary information first must be specified by the user then embed in the unit to form the required interconnection pattern.

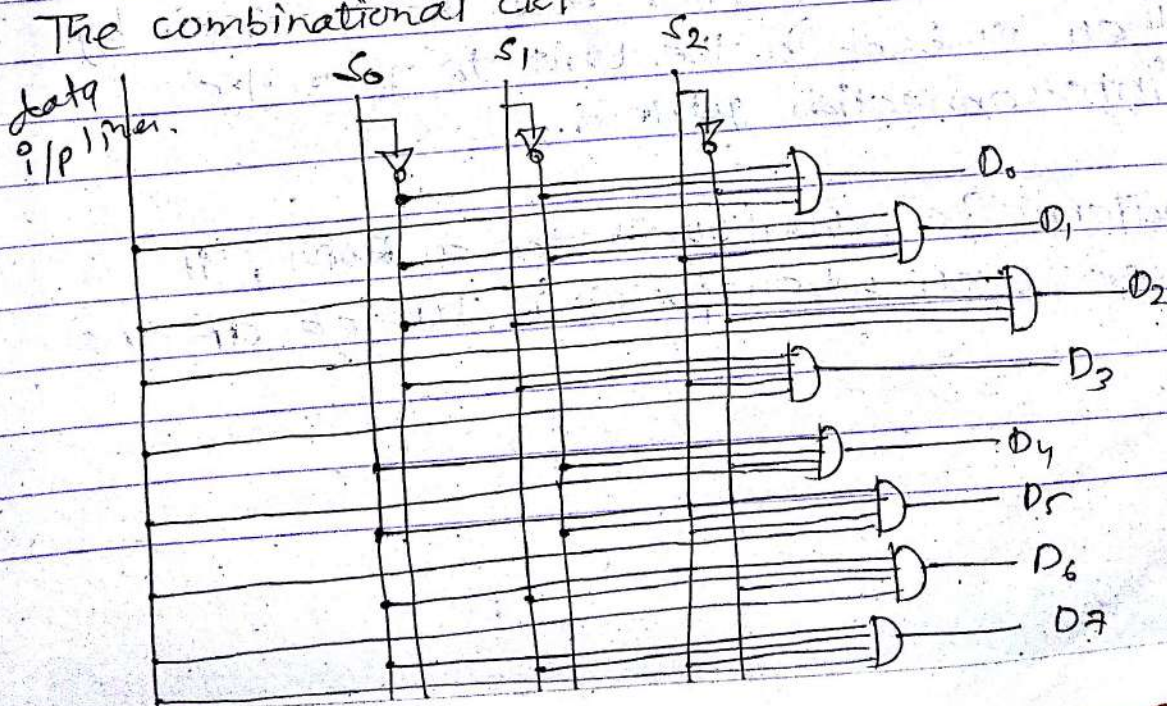
↳ Once a pattern is stabilised for a ROM, it remains fixed even when power is turned off and ON again.

2) 1:8 demultiplexer:

It has one input line, 8 output lines and 3 selection lines. Suppose S_0, S_1, S_2 be the selection lines and $D_0, D_1, D_2, D_3, D_4, D_5, D_6, D_7$ be the o/p lines. Now truth table or functional table for 1:8 demultiplexer is

| Selection lines | | | O/P | | | | | | | |
|-----------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| S_0 | S_1 | S_2 | D_0 | D_1 | D_2 | D_3 | D_4 | D_5 | D_6 | D_7 |
| 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| 0 | 1 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 |
| 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 |
| 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 |
| 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |

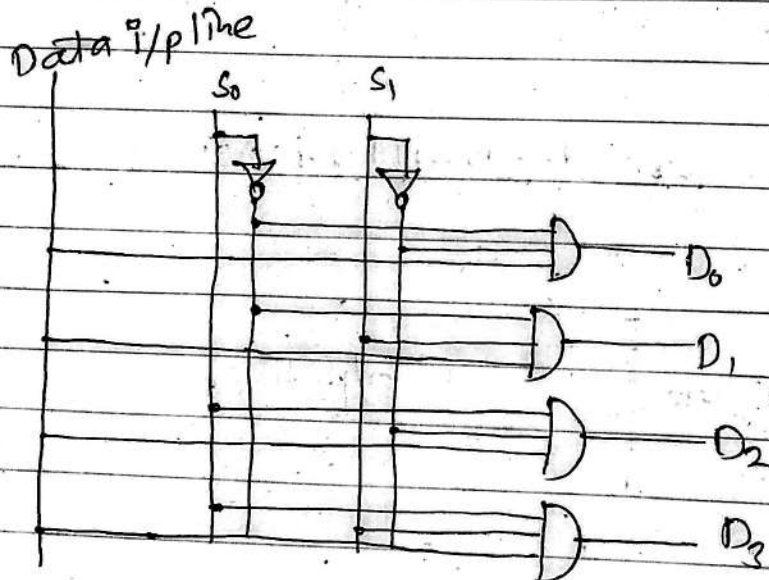
The combinational ckt



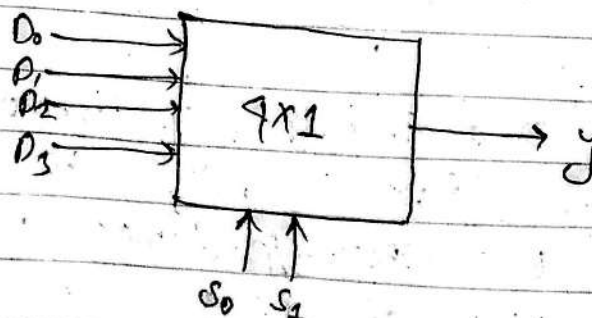
DB box:

| Selection lines | | O/p | | | |
|-----------------|-------|-------|-------|-------|-------|
| S_0 | S_1 | D_0 | D_1 | D_2 | D_3 |
| 0 | 0 | 1 | 0 | 0 | 0 |
| 0 | 1 | 0 | 1 | 0 | 0 |
| 1 | 0 | 0 | 0 | 1 | 0 |
| 1 | 1 | 0 | 0 | 0 | 1 |

The combinational circuit



Block diagram:



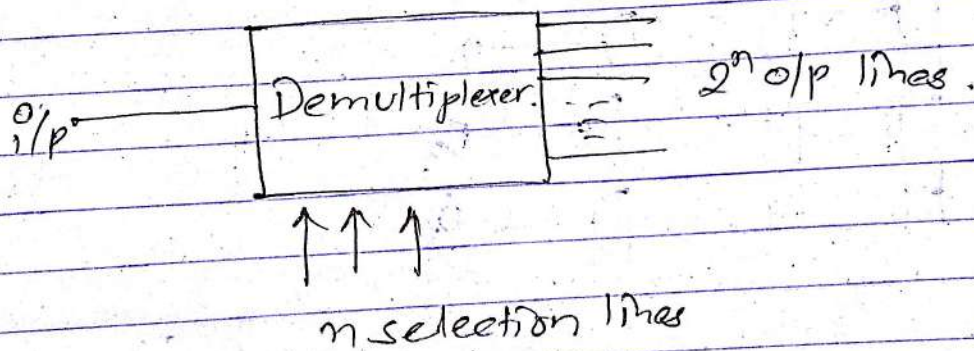
De-multiplexer:

→ De-multiplexer means one to many. It is a combinational logic circuit with one input and many o/p. along with same control signal.

→ It takes a data from a signal line and distributes them to a given no. of output lines.

→ If the information is 1 of 2^n possible output with n selection lines.

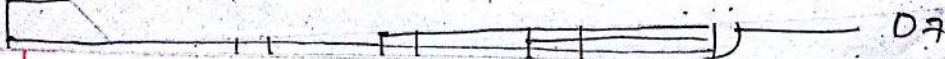
→ The block diagram of de-multiplexer is



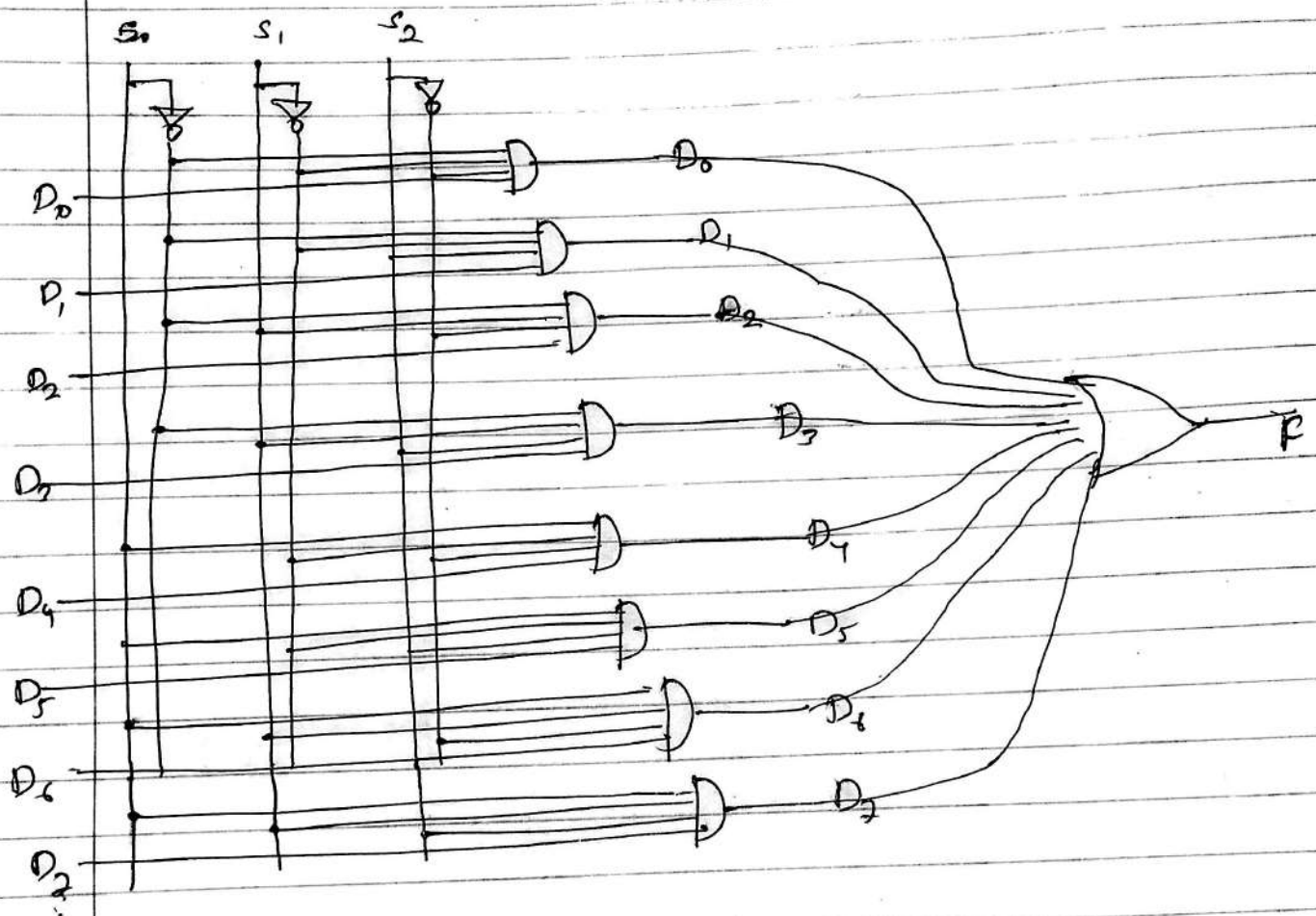
for example:

→ 1:4 demultiplexer:

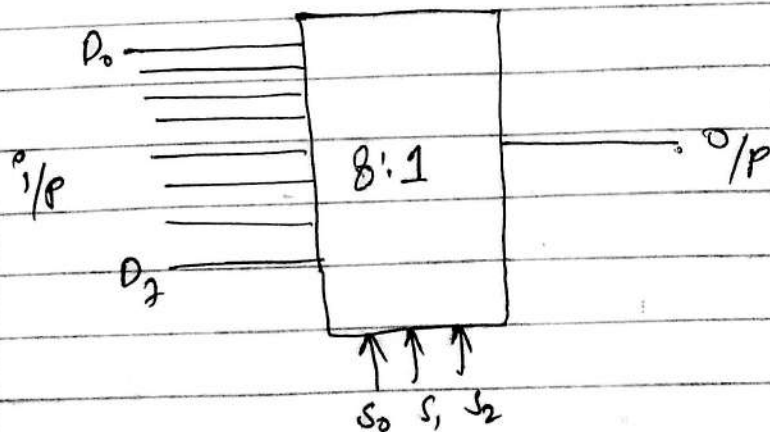
It has one input, 4 output and two selection lines. Suppose S_0, S_1 be the select lines and D_0, D_1, D_2 be the o/p lines. Now truth table or functional table of 1:4 demultiplexer



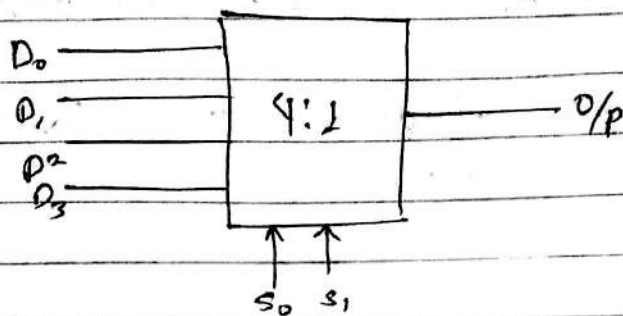
→ Combinationed ckt diagram:



Block Diagram:



Block diagram:



(ii) 8×1 multiplexer:

→ In $8:1$ multiplexer 8 i/p lines, 3 selection lines and 1 O/p lines are present.

→ let 8 i/p lines are denoted by $D_0, D_1, D_2, D_3, D_4, D_5, D_6, D_7$ and three selection lines are S_0, S_1, S_2 .

→ the required logic diagram, function table and block diagram is:

function table:

| selection lines | | | o/p |
|-----------------|-------|-------|-------|
| S_0 | S_1 | S_2 | |
| 0 | 0 | 0 | D_0 |
| 0 | 0 | 1 | D_1 |
| 0 | 1 | 0 | D_2 |
| 0 | 1 | 1 | D_3 |
| 1 | 0 | 0 | D_4 |
| 1 | 0 | 1 | D_5 |
| 1 | 1 | 0 | D_6 |
| 1 | 1 | 1 | D_7 |

- Let 4 i/p lines are denoted by D_0, D_1, D_2 & D_3 and two selection lines are S_0 & S_1 .
- the required logic diagram, function table and block diagram is shown.

function table:

| selection line | | O/p |
|----------------|-------|-------|
| S_0 | S_1 | F |
| 0 | 0 | D_0 |
| 0 | 1 | D_1 |
| 1 | 0 | D_2 |
| 1 | 1 | D_3 |

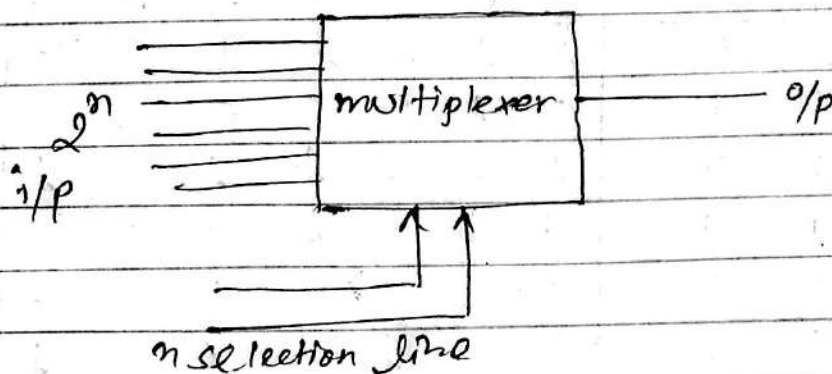
$$i/p \ 2^n \xrightarrow{\text{selection line } n} o/p$$

Multiplexer:

Multiplexing means transmitting a large number of information units ~~or~~ over a smaller no. of channel.

It means many input and single output (many-to-one). A multiplexer is a combinational circuit that selects binary information from several i/p lines and directs it to a signal by a set of selection lines.

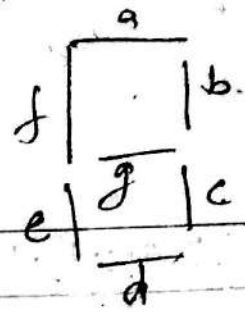
Normally there are 2^n i/p lines, n selection line and one output.



Types:

① 4X1 multiplexer:

— In 4:1 multiplexer 4 input line & 2 selection line and 1 o/p lines are present.



BCD to 7-segment Decoder:

Multiplexing means transmitting a lot of data over a single line.

A BCD to 7-segment decoder is a combinational ckt that accepts a decimal digit in BCD and generates appropriate output. The 7-segment output of decoder is a, b, c, d, e, f, g, and BCD input w, x, y, z. The corresponding segment in the display as shown in fig.

The truth table for BCD to 7-segment.

| Input | Input | | | | Output | | | | | | |
|-------|-------|---|---|---|--------|---|---|---|---|---|---|
| | w | x | y | z | a | b | c | d | e | f | g |
| 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 0 |
| 1 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| 2 | 0 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 1 |
| 3 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 1 |
| 4 | 0 | 1 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 |
| 5 | 0 | 1 | 0 | 1 | 1 | 0 | 1 | 1 | 0 | 1 | 1 |
| 6 | 0 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 |
| 7 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 |
| 8 | 1 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 9 | 1 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 1 |

may be wrong.

g

-the circuit diagram for decimal to BCD encoder is :



| Input | | | | | | | | | | Output | | | |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--------|---|---|---|
| D_0 | D_1 | D_2 | D_3 | D_4 | D_5 | D_6 | D_7 | D_8 | D_9 | D | C | B | A |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 |
| 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 2 |
| 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 3 |
| 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 4 |
| 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 5 |
| 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 1 | 6 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 1 | 7 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 8 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 9 |

The canonical form for A o/p

$$A = \Sigma(1, 3, 5, 7, 9)$$

$$= D_1 + D_3 + D_5 + D_7 + D_9$$

1/2

~~for B,~~

$$B = \Sigma(2, 3, 6, 7)$$

$$= D_2 + D_3 + D_6 + D_7$$

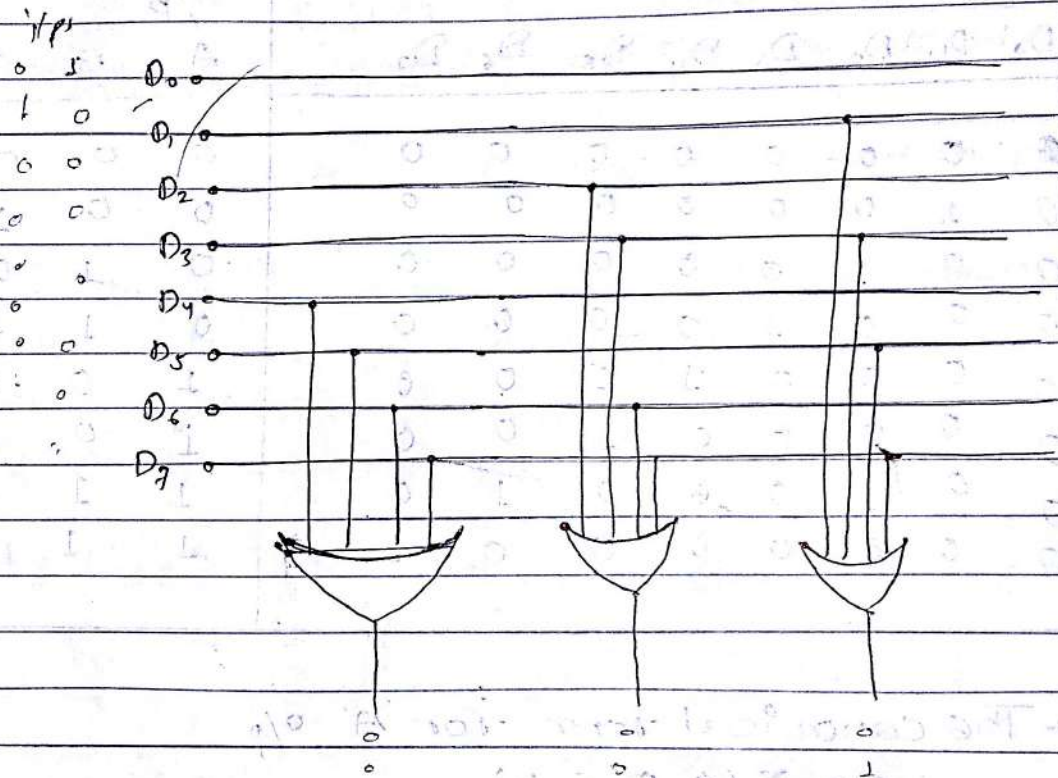
$$C = \Sigma(4, 5, 6, 7)$$

$$= D_4 + D_5 + D_6 + D_7$$

$$D = \Sigma(8, 9)$$

$$= D_8 + D_9$$

- The circuit diagram of octal to binary encoder



(2.) Design a combinational logic ckt for encoder to convert decimal no. to BCD.
Solution:

$$BCD = 2D + 2D + 2D + 2D$$

$$BCD = 2D + 2D + 2D + 2D$$

$$BCD = 2D + 2D + 2D + 2D$$

$$BCD = 2D + 2D + 2D + 2D$$

Truth table:

| I/p | | | | | | | | O/p | | |
|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|-----|---|---|
| D ₀ | D ₁ | D ₂ | D ₃ | D ₄ | D ₅ | D ₆ | D ₇ | A | B | C |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |
| 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 |
| 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 |
| 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 1 |
| 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 1 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 |

- The canonical form for 'A' o/p

$$A = \sum(4, 5, 6, 7)$$

$$= D_4 + D_5 + D_6 + D_7$$

- The canonical form for 'B' o/p

$$B = \sum(2, 3, 6, 7)$$

$$= D_2 + D_3 + D_6 + D_7$$

- The canonical form for 'C' o/p

$$C = \sum(1, 2, 5, 7)$$

$$= D_1 + D_2 + D_5 + D_7$$

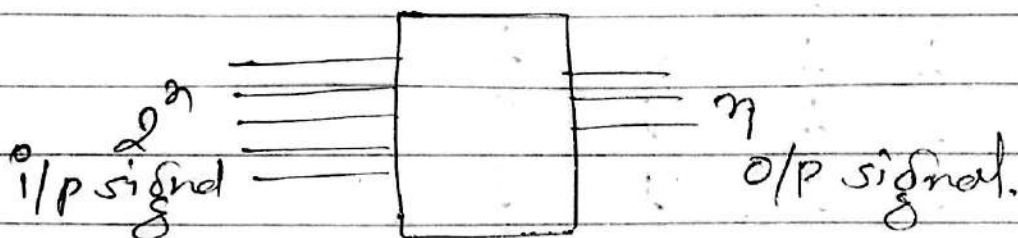
Encoder

An encoder is a digital circuit that performs the inverse operation of decoder.

it has 2^n input lines and n output lines

this device is useful to convert the decimal digit into binary digit, where an encoder converts active input signal into a coded o/p signal.

the block diagram of encoder is



Questions

Design a combinational ckt which gives octal to binary encoder output.

Solution:

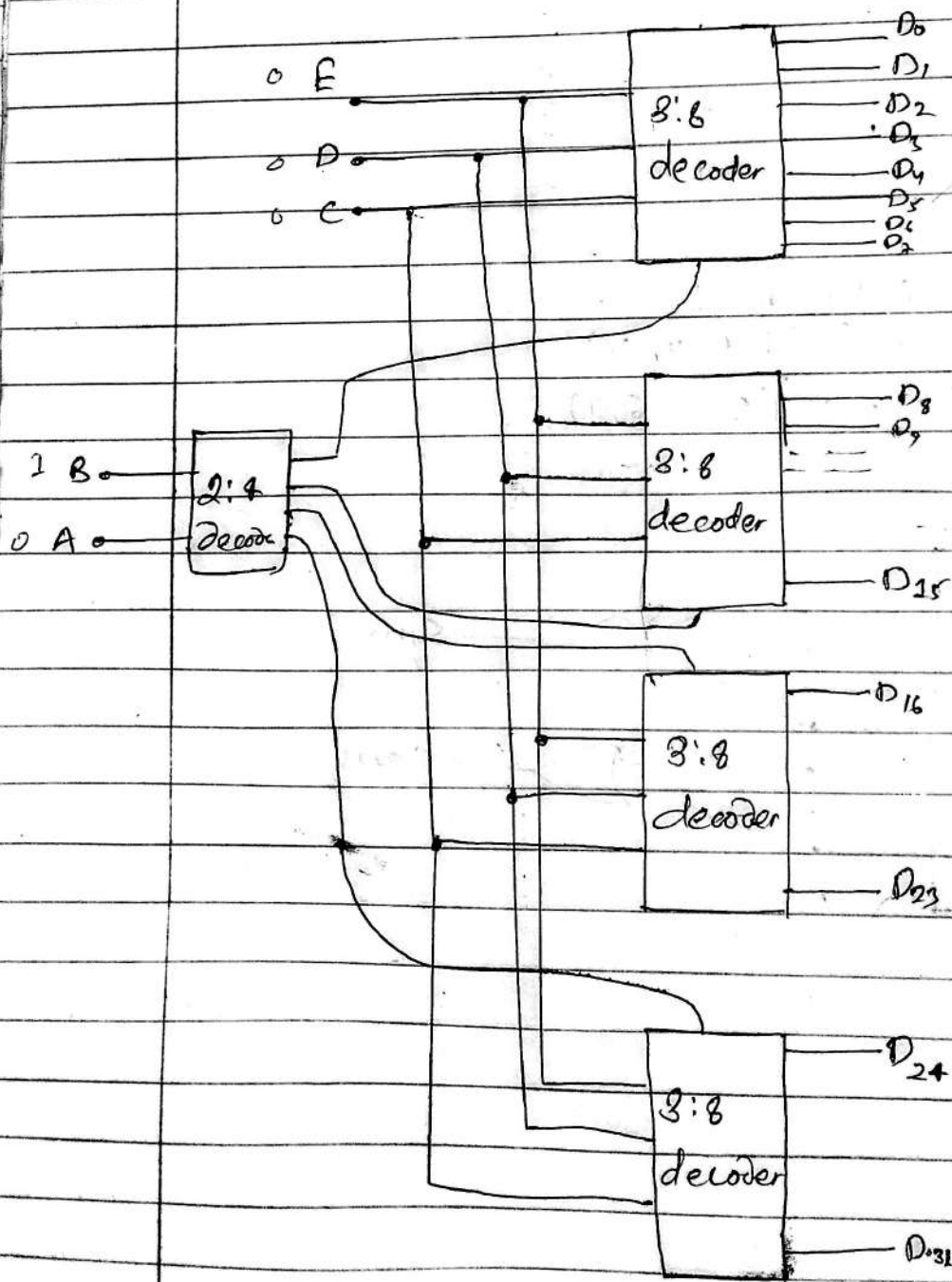
Let us consider the i/p variable are $D_0, D_1, \dots, D_7$

| | Input | | | | | |
|----------------|-------|---|---|---|---|---|
| A | 0 | 0 | 0 | 0 | 0 | F |
| B | 0 | 0 | 0 | 0 | 0 | D |
| C | 0 | 0 | 0 | 0 | 0 | E |
| D ₀ | 0 | 0 | 0 | 0 | 0 | |
| D ₁ | 0 | 0 | 0 | 0 | 0 | |
| D ₂ | 0 | 0 | 0 | 0 | 0 | |
| D ₃ | 0 | 0 | 0 | 0 | 0 | |

① Implement 5:32 (5X32) decoder using four 8:8 decoder and one 2:4 decoder.

solution:

4 $\rightarrow$ 8:8 decoder
1 $\rightarrow$ 2:4 decoder.



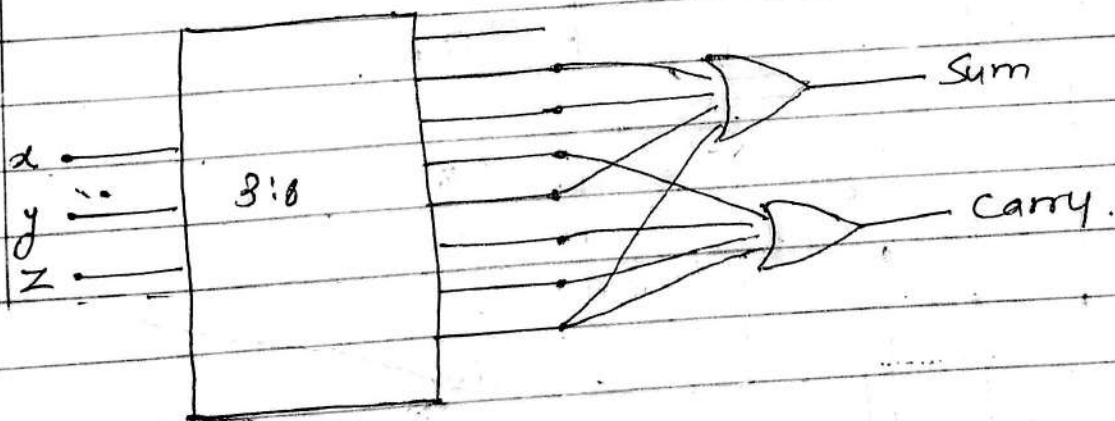
Q.N.2. solution:

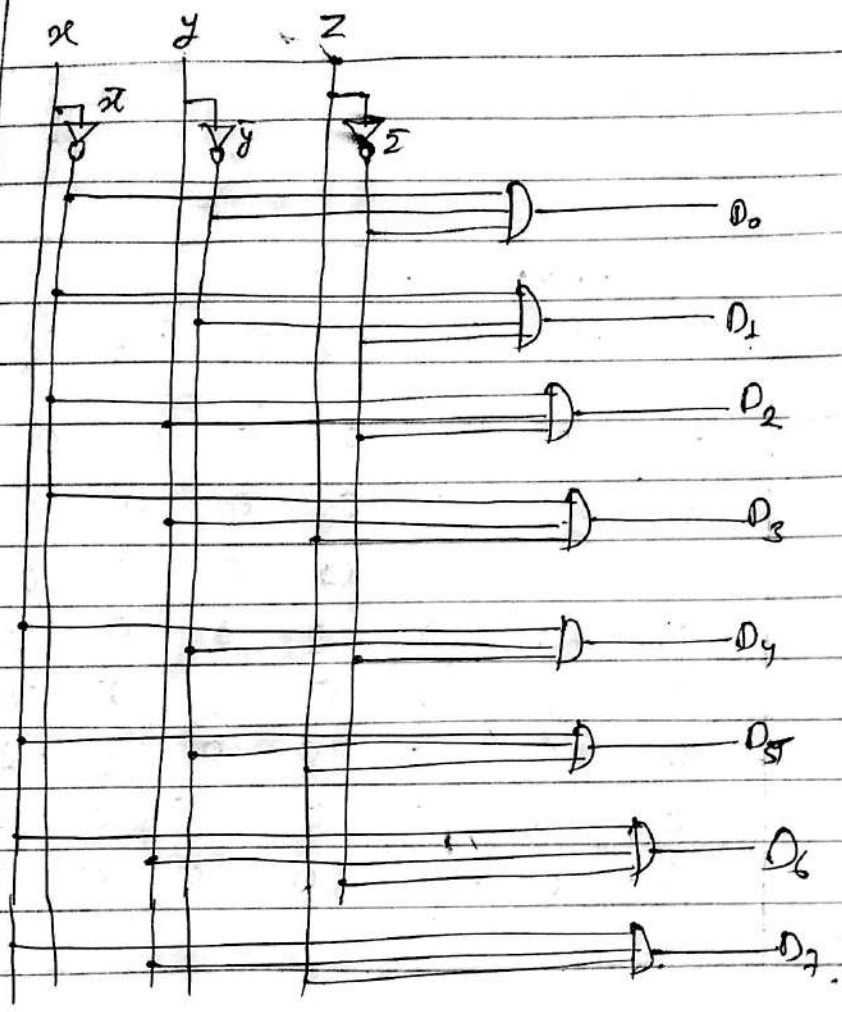
| Input | | | Output | |
|-------|---|---|--------|-------|
| x | y | z | Sum | Carry |
| 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 | 0 |
| 0 | 1 | 0 | 1 | 0 |
| 0 | 1 | 1 | 0 | 1 |
| 1 | 0 | 0 | 1 | 0 |
| 1 | 0 | 1 | 0 | 1 |
| 1 | 1 | 0 | 0 | 1 |
| 1 | 1 | 1 | 1 | 1 |

Here, canonical form for:

$$\text{sum (s)} = \Sigma(1, 2, 4, 7)$$

$$\text{Carry (c)} = \Sigma(3, 5, 6, 7)$$





b. 3:8 decoder.

Truth table:

| Input | | | Output lines | | | | | | | |
|-------|---|---|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| x | y | z | D ₀ | D ₁ | D ₂ | D ₃ | D ₄ | D ₅ | D ₆ | D ₇ |
| 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| 0 | 1 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 |
| 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 |
| 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 |
| 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |

Boolean function:

$$D_0 = \bar{x} \bar{y} \bar{z}$$

$$D_1 = \bar{x} \bar{y} z$$

$$D_2 = \bar{x} y \bar{z}$$

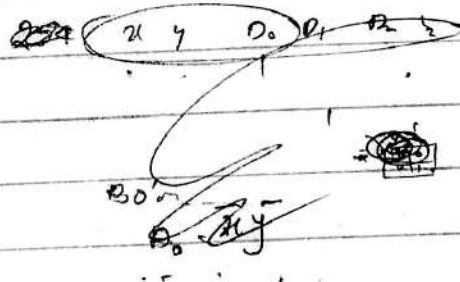
$$D_3 = \bar{x} y z$$

$$D_4 = x \bar{y} \bar{z}$$

$$D_5 = x \bar{y} z$$

$$D_6 = x y \bar{z}$$

$$D_7 = x y z$$



| I/p | | Output lines | | | |
|-----|---|----------------|----------------|----------------|----------------|
| x | y | D ₀ | D ₁ | D ₂ | D ₃ |
| 0 | 0 | 1 | 0 | 0 | 0 |
| 0 | 1 | 0 | 1 | 0 | 0 |
| 1 | 0 | 0 | 0 | 1 | 0 |
| 1 | 1 | 0 | 0 | 0 | 1 |

Boolean function:

$$D_0 = \bar{x}\bar{y}$$

$$D_1 = \bar{x}y$$

$$D_2 = x\bar{y}$$

$$D_3 = xy$$

Logical circuit diagram of decoder:

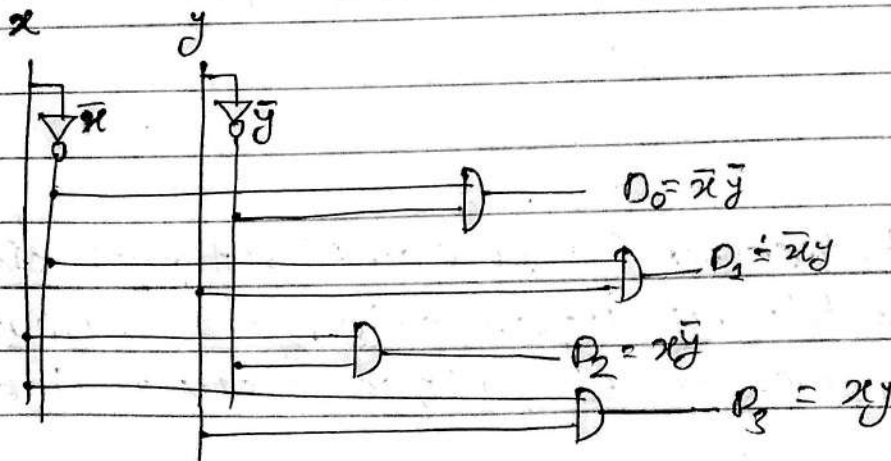


fig: CRT of decoder.

Chapter-6

MSI and LSI components in CLC or diagram.

Integration:

The circuit complexity or integrated ckt have been classified under 4 level of integration, namely Small scale Integration (SSI). Example: Logical gates.

II. MSI (Medium scale integration.)
eg: adder, subtractor, decoder, encoder.

III. LSI (Large scale integration).
ALU, microcontroller, microprocessor etc.

IV. VLSI (Very large scale integration)
Programmable integrated ckt (PIC)

Decoder:

A decoder is a combinational circuit that converts binary information from input line to a maximum no. of 2^n unique output lines. for example:

a. 2:4 decoder

Truth-table:

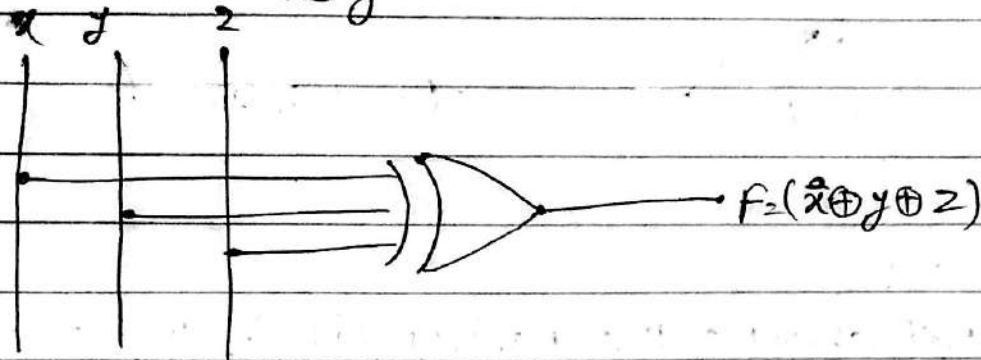


OR

| $x \backslash yz$ | $\bar{y}z$ | $\bar{y}\bar{z}$ | yz | $y\bar{z}$ |
|-------------------|------------|------------------|------|------------|
| $\bar{x}$ | 1 | 0 | 1 | 0 |
| x | 0 | 1 | 0 | 1 |

$$f = \bar{x}\bar{y}z + x\bar{y}z + \bar{x}yz + xy\bar{z}$$

$$f = x \oplus y \oplus z$$



Even parity checker:

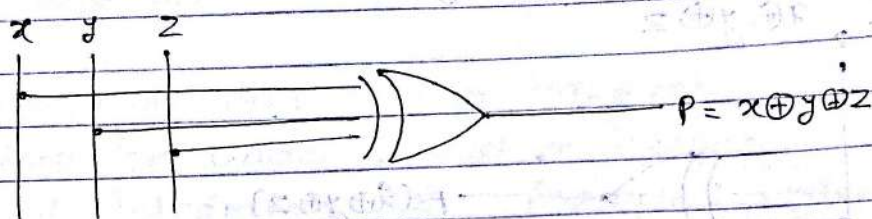
In even parity checker information must be transmitted with even parity, the four bit data (message) received must have an even no of 1's. An error occurred during the transmission of four bit message received as an odd number of 1's indicating that one bit has changed the value during the transmission.

The parity bit detects the presence of any odd combination of errors and even combination of errors is undetectable.

K-map for P:

| | | | | |
|-----------|------------|------------|------|------------|
| | $\bar{y}z$ | $y\bar{z}$ | yz | $y\bar{z}$ |
| $\bar{x}$ | 0 | 1 | 0 | 1 |
| x | 1 | 0 | 1 | 0 |

$$\begin{aligned}
 \therefore P &= \bar{x}\bar{y}z + \bar{x}y\bar{z} + x y z + x \bar{y}\bar{z} \\
 &= \bar{x}(\bar{y}z + y\bar{z}) + x(yz + \bar{y}\bar{z}) \\
 &= \bar{x}(y \oplus z) + x(y \oplus z) \\
 &= x \oplus y \oplus z
 \end{aligned}$$



Parity checker:

The circuits that check the parity bit in the receiver is called the parity checker.

from the truth table the odd parity checker is

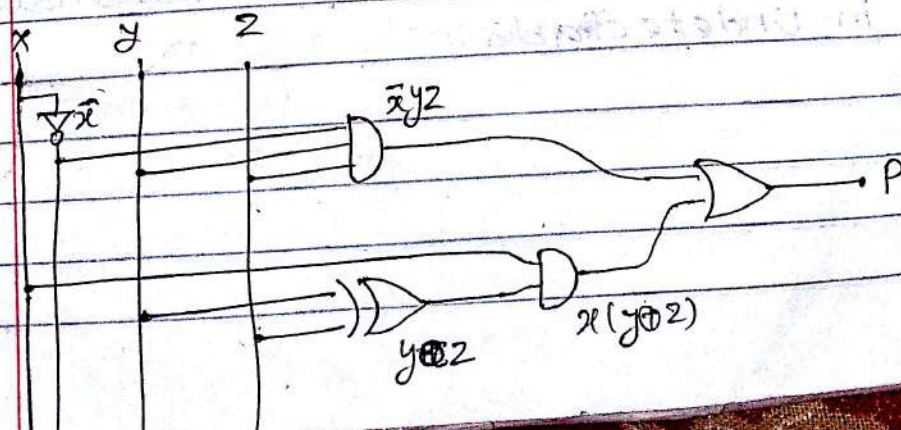
| x | y | z | P |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 |
| 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | 0 |
| 1 | 0 | 0 | 1 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 0 |
| 1 | 1 | 1 | 1 |

? (why)

K-map for P:

| | | | | |
|-----------|------------|------------|------|------------|
| | $\bar{y}z$ | $y\bar{z}$ | yz | $y\bar{z}$ |
| $\bar{x}$ | 1 | 0 | 1 | 0 |
| x | 0 | 1 | 0 | 1 |

$$\begin{aligned}
 \therefore P &= \bar{x}\bar{y}z + \bar{x}y\bar{z} + x y z \\
 &= \bar{x}y z + x(\bar{y}z + y\bar{z}) \\
 &= \bar{x}y z + x(y \oplus z)
 \end{aligned}$$



Parity generation and checking:

A parity bit is an extra bit included with a binary message to make number of 1's either odd or even then the message including the parity bit is transmitted and checked at the receiving end for the errors.

An error is detected if the checked parity doesn't respond with the one transmitted.

Parity generator:

The circuits that generates the parity bit in the transmitter is called parity generator.

eg: Consider a three bit message to be transmitted together with an even parity bit. for even parity 'P' must be generated to make the total no. of 1's even.

from truth table - 'P' contains an odd function.

i.e. $P = 1$.

The truth table is:

| x | y | z | P |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 |
| 0 | 1 | 0 | 1 |

- Parity generation and checking:

A parity bit is an extra bit included with a binary message to make number of 1's either odd or even then the message including the parity bit is transmitted and checked at the receiving end for the errors.

An error is detected if the checked parity doesn't respond with the one transmitted.

Parity generator:

The circuits that generates the parity bit in the transmitter is called parity generator.

eg: Consider a three bit message to be transmitted together with an even parity bit. for even parity 'P' must be generated to make the total no. of 1's even.

from truth table - 'P' contains an odd function.

ie, $P = 1$.

The truth table is:

| x | y | z | P |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 |
| 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | 0 |
| 1 | 0 | 0 | 1 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 0 |
| 1 | 1 | 1 | 1 |

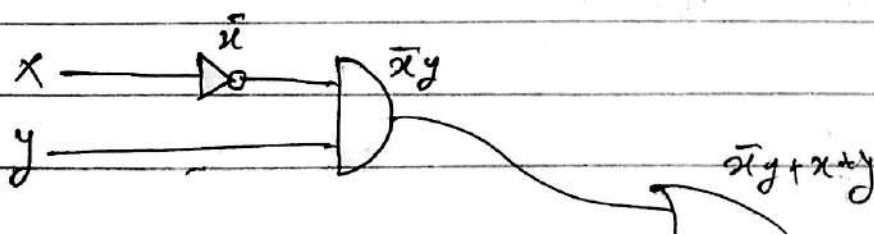
MULTILEVEL NAND & NOR GATE

→ MULTILEVEL NAND and NOR gate are defined as the combination of one and only NAND gate OR NOR gate.

→ for multilevel NAND and NOR gate we use universal gate called NAND gate and NOR gate.

→ for multilevel NAND and NOR gate, we replace different basic gates by using universality of NAND and NOR gate. Similarly we replace NOT gate, AND gate and OR gate by the combination of either NAND gate or NOR gate.

→ for the following circuitary make multilevel NAND gate and NOR gate.



. NAND .

1. NOT
2. AND
3. OR

- the k-map for difference (D) is

| $x \backslash yz$ | $\bar{y}z$ | $y\bar{z}$ | yz | $\bar{y}\bar{z}$ |
|-------------------|------------|------------|------|------------------|
| $\bar{x}$ | 0 | 1 | 0 | 1 |
| x | 1 | 0 | 1 | 0 |

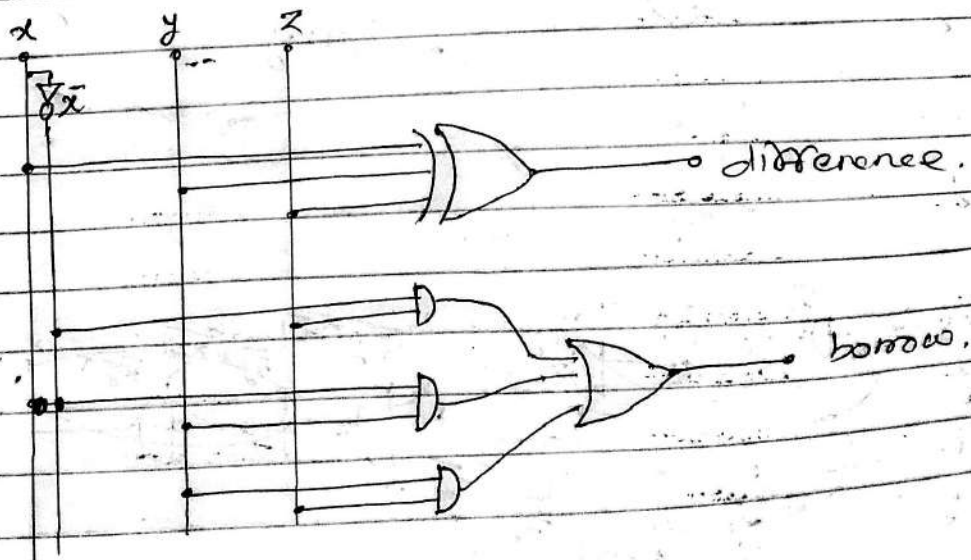
$$\begin{aligned}
 \therefore F_D &= \bar{x}\bar{y}z + \bar{x}y\bar{z} + x\bar{y}\bar{z} + xyz \\
 &= z(\bar{x}\bar{y} + x\bar{y}) + \bar{z}(\bar{x}y + xy) \\
 &= z(\bar{x} \oplus y) + \bar{z}(x \oplus y) \\
 &= x \oplus y \oplus z
 \end{aligned}$$

- the k-map for borrow (B) is

| $x \backslash yz$ | $\bar{y}z$ | $y\bar{z}$ | yz | $\bar{y}\bar{z}$ |
|-------------------|------------|------------|------|------------------|
| $\bar{x}$ | 0 | 1 | 1 | 1 |
| x | 0 | 0 | 1 | 0 |

$$\therefore F_B = \bar{x}z + \bar{x}y + yz$$

- the combinational logic circuit (CLC) of full subtractor is



- full subtractor:

* Full subtractor is used for higher ordered column is a logic circuit that can subtract three bit at a time, hence a full subtractor is a combinational logic circuit that performs arithmetic subtraction of three bit input and produce difference and borrow at the output.

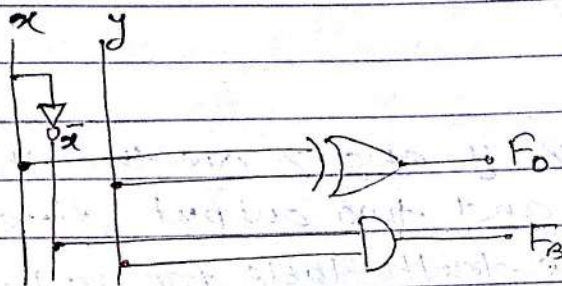
* Let x , y , and z are the three inputs of full subtractor and two output difference (D) and borrow (B). The truth table for full subtractor is

| Input | Output |
|-------|--------|
| | |

| $x \backslash y$ | 0 | 1 |
|------------------|---|---|
| $\bar{x}$ | 0 | 1 |
| x | 0 | 0 |

$$\therefore f_b = \bar{x}y$$

the combinational circuit for half subtractor is:



- difference and borrow can be performed using two bit by the logic circuit called half subtractor.
- the output of exclusive OR gate is called difference while output of AND gate is called borrow.
- the table above shows the truth table of half subtractor and hence half subtraction subtractor performs binary subtraction. It does electrically what we do manually.

- half subtractor:

It is a combinational logic circuit with two input x and y and two output difference (D) and borrow (B) . It is a basic circuit for subtraction of two single bit no.

design:

1. Subtraction of two bit
2. two input and two output
3. let two inputs are x & y and o/p difference (D) and borrow (B) .
4. The truth table for half subtractor is

| Input | | Output | |
|-------|-----|------------------|--------------|
| x | y | difference (D) | Borrow (B) |
| 0 | 0 | 0 | 0 |
| 0 | 1 | 1 | 1 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 0 |

$$\therefore f_c = yz + xz + xy$$

— The k-map for sum (s)

| $x \backslash yz$ | $\bar{y}\bar{z}$ | $\bar{y}z$ | $y\bar{z}$ | yz |
|-------------------|------------------|------------|------------|------|
| $\bar{x}$ | 0 | 1 | 0 | 1 |
| x | 1 | 0 | 1 | 0 |

$$\therefore f_s = \bar{x}\bar{y}z + \bar{x}y\bar{z} + x\bar{y}\bar{z} + xy\bar{z}$$

$$= x \oplus y \oplus z$$

$$f_s = \bar{x}\bar{y}z + x\bar{y}z + \bar{x}y\bar{z} + xy\bar{z}$$

$$= z(\bar{x}\bar{y} + xy) + \bar{z}(\bar{x}y + x\bar{y})$$

$$= z(\bar{x} \oplus y) + \bar{z}(x \oplus y)$$

$$(x \oplus y \oplus z)$$

— The combinational logic circuit (CLC) of following full adder is

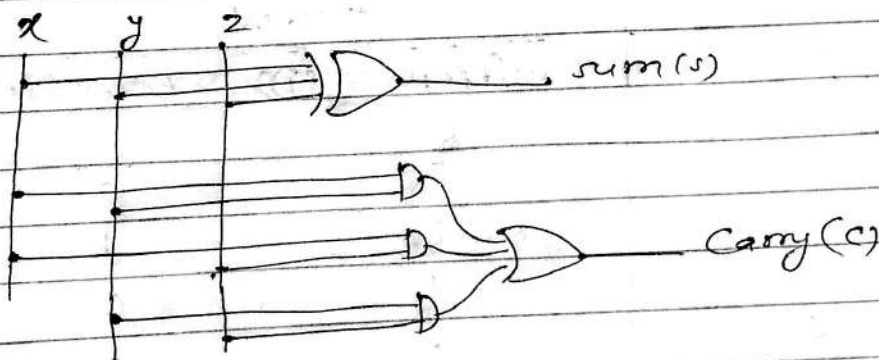


fig: CLC of full adder.

SUBTRACTOR:

Subtractor is a combinational ckt that performs subtraction of binary digits. There are two types of subtractor.

- half subtractor
- full subtractor.

2. full adder:

- Full adder is used for higher order column is a logic circuit that can add three bit at a time. hence a full adder is a combinational logic ckt that performs arithmetic sum of three bit input and produce sum and carry at the o/p.

- let x, y, z are three inputs of full adder and two o/p: sum and carry (C). The truth table of full adder is

| Input | | | Output | |
|-------|-----|-----|---------|-----------|
| x | y | z | sum (s) | Carry (c) |
| 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 | 0 |
| 0 | 1 | 0 | 1 | 0 |
| 0 | 1 | 1 | 0 | 1 |
| 1 | 0 | 0 | 1 | 0 |
| 1 | 0 | 1 | 0 | 1 |
| 1 | 1 | 0 | 0 | 1 |
| 1 | 1 | 1 | 1 | 1 |

- The K-map for sum (s) carry (c)

| | | | | |
|-----------|-------------------|-------------------|-------------|-------------------|
| | $\bar{x}\bar{y}z$ | $\bar{x}y\bar{z}$ | $\bar{x}yz$ | $x\bar{y}\bar{z}$ |
| $\bar{x}$ | 0 | 0 | 1 | 0 |
| x | 0 | 1 | 0 | 1 |

- The k-map for sum (s).

| $x \backslash y$ | $\bar{y}$ | y |
|------------------|-----------|-----|
| $\bar{x}$ | 0 | 1 |
| x | 1 | 0 |

$$f_s = x\bar{y} + \bar{x}y$$

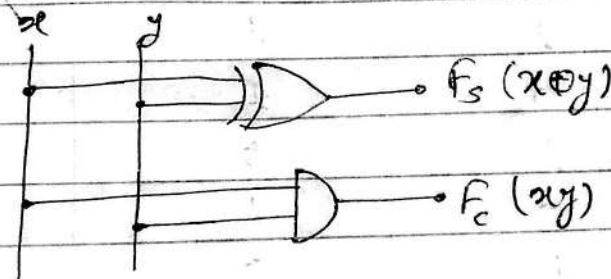
$$f_s = x \oplus y$$

- k-map for carry (c)

| $x \backslash y$ | $\bar{y}$ | y |
|------------------|-----------|-----|
| $\bar{x}$ | 0 | 0 |
| x | 0 | 1 |

$$f_c = xy$$

- The combination ckt of half adder is



- Sum and carry can be performed using two bit by the logic ckt called half adder.

- The o/p of exclusive OR gate is called sum and while o/p of AND gate is called carry (c).

- The table above shows the truth table of half adder hence half adder performs binary addition. It does electrically what we do manually.

ADDER

An adder is a combinational ckt that performs addition of binary digits. There are two types of adder.

1. Half adder
2. Full adder.

1. Half adder:

It is a combinational logic ckt with two input x and y and two output sum (s) and carry (c). It is basic ckt for addition of two signal bit no.

design:

1. Addition of two bit.
2. Two input and two output
3. let two inputs are x and y and output sum (s) and carry (c).
4. The truth table of half adder is

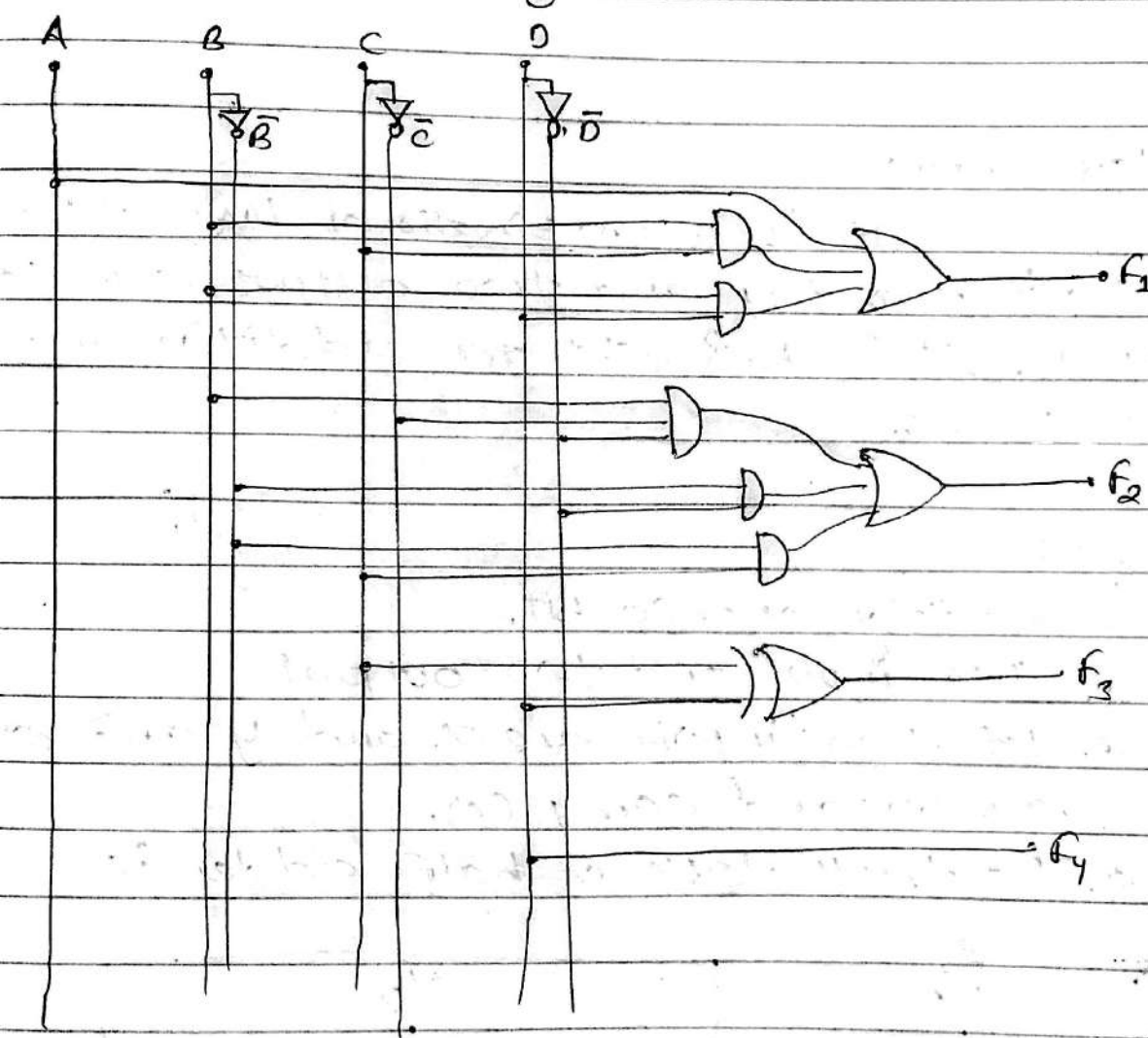
| i/p | | o/p | |
|-----|-----|-------------|---------------|
| x | y | sum (s) | Carry (c) |
| 0 | 0 | 0 | 0 |
| 0 | 1 | 1 | 0 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 1 |

- the k-map for f_4 is

| | $\bar{C}D$ | $C\bar{D}$ | CD | $\bar{C}\bar{D}$ |
|------------------|------------|------------|------|------------------|
| $\bar{A}\bar{B}$ | 0 | 1 | 1 | 0 |
| $\bar{A}B$ | 0 | 1 | 1 | 0 |
| AB | X | X | X | X |
| $A\bar{B}$ | 0 | 1 | X | X |

$\therefore f_4 = D$

- the combinational logic ckt of above function is



8. $\begin{matrix} 1 & 1 & 1 & 1 \\ 2 & 4 & 2 & 1 \\ 1 & 1 & 1 & 1 \end{matrix} \rightarrow \text{so } (8, 4, -2, -1) \text{ ની ઉપર 9 કોષ્ટકો છે. 4 કોષ્ટકો 2 કોષ્ટકો ની ઉપર 4 કોષ્ટકો ની}$
 $\begin{matrix} 8 & 4 & -2 & -1 \\ 1 & 1 & 1 & 1 \end{matrix}$

| | 2 | 4 | 2 | 1 | | 8 | 4 | -2 | -1 |
|---|---|---|---|---|--|-------|-------|-------|-------|
| | A | B | C | D | | f_1 | f_2 | f_3 | f_4 |
| 0 | 0 | 0 | 0 | 0 | | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 1 | | 0 | 1 | 1 | 1 |
| 2 | 0 | 0 | 1 | 0 | | 0 | 1 | 1 | 0 |
| 3 | 0 | 0 | 1 | 1 | | 0 | 1 | 0 | 1 |
| 4 | 0 | 1 | 0 | 0 | | 0 | 1 | 0 | 0 |
| 5 | 0 | 1 | 0 | 1 | | 1 | 0 | 1 | 1 |
| 6 | 0 | 1 | 1 | 0 | | 1 | 0 | 1 | 0 |
| 7 | 0 | 1 | 1 | 1 | | 1 | 0 | 0 | 1 |
| 8 | 1 | 0 | 0 | 0 | | 1 | 0 | 0 | 0 |
| 9 | 1 | 1 | 1 | 1 | | 1 | 1 | 1 | 1 |

the k-map for f_1 :

| AB \ CD | 00 | 01 | 11 | 10 |
|------------------|----|----|----|----|
| $\bar{A}\bar{B}$ | 0 | 0 | 0 | 0 |
| $\bar{A}B$ | 0 | 1 | 1 | 1 |
| AB | 1 | 1 | 1 | 1 |
| $A\bar{B}$ | 1 | 1 | 1 | 1 |

$$\therefore f_1 = A + BC + BD$$

the k-map for f_2 :

| AB \ CD | 00 | 01 | 11 | 10 |
|------------------|----|----|----|----|
| $\bar{A}\bar{B}$ | 0 | 1 | 1 | 1 |
| $\bar{A}B$ | 1 | 0 | 0 | 0 |
| AB | 1 | 1 | 1 | 1 |
| $A\bar{B}$ | 1 | 1 | 1 | 1 |

$$\therefore f_2 = B\bar{C}\bar{D} + \bar{B}D + \bar{B}C$$

the k-map for f_3 :

| AB \ CD | 00 | 01 | 11 | 10 |
|------------------|----|----|----|----|
| $\bar{A}\bar{B}$ | 0 | 1 | 0 | 1 |
| $\bar{A}B$ | 0 | 1 | 0 | 1 |
| AB | 1 | 1 | 1 | 1 |
| $A\bar{B}$ | 1 | 1 | 1 | 1 |

$$\therefore f_3 = \bar{C}D + C\bar{D}$$

$$= C \oplus D$$

- the k-map for f_3 :

| $AB \backslash CD$ | $\bar{C}\bar{D}$ | $\bar{C}D$ | CD | $C\bar{D}$ |
|--------------------|------------------|------------|------|------------|
| $\bar{A}\bar{B}$ | 0 | 0 | 0 | 0 |
| $\bar{A}B$ | 1 | 1 | 1 | 1 |
| AB | 0 | X | X | X |
| $A\bar{B}$ | 0 | 0 | 0 | 0 |

$$\therefore f_3 = \bar{A}B$$

- the k-map of f_4 :

| $AB \backslash CD$ | $\bar{C}\bar{D}$ | $\bar{C}D$ | CD | $C\bar{D}$ |
|--------------------|------------------|------------|------|------------|
| $\bar{A}\bar{B}$ | 0 | 0 | 1 | 1 |
| $\bar{A}B$ | 0 | 0 | 1 | 1 |
| AB | 1 | X | X | X |
| $A\bar{B}$ | 0 | 0 | 0 | 0 |

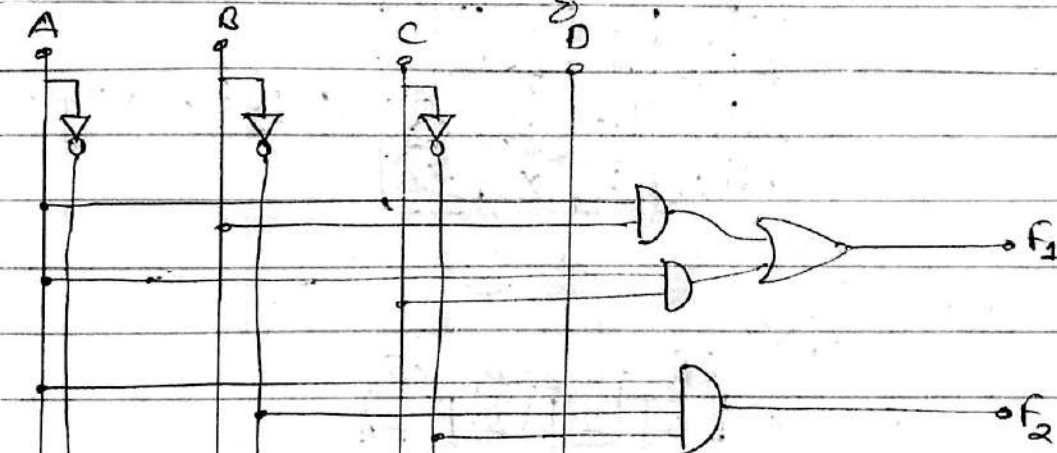
$$\therefore f_4 = \bar{A}C + AB$$

- the k-map for f_5 :

| $AB \backslash CD$ | $\bar{C}\bar{D}$ | $\bar{C}D$ | CD | $C\bar{D}$ |
|--------------------|------------------|------------|------|------------|
| $\bar{A}\bar{B}$ | 0 | 1 | 1 | 0 |
| $\bar{A}B$ | 0 | 1 | 1 | 0 |
| AB | 0 | X | X | X |
| $A\bar{B}$ | 0 | 1 | 1 | 0 |

$$\therefore f_5 = D$$

- the combinational logic circuit for above function is



2. $84-2-1 \rightarrow BCD$
 $\frac{1 \ 1 \ 0 \ 0}{4 \ 6H} = 12$
 $12 = 00010010 \rightarrow 54H$

| | 8 | 4 | 2 | 1 | BCD |
|----|---|---|---|---|-------------------------------|
| | A | B | C | D | $F_1 \ F_2 \ F_3 \ F_4 \ F_5$ |
| 0 | 0 | 0 | 0 | 0 | 0 0 0 0 0 |
| 1 | 0 | 1 | 1 | 1 | 0 0 0 0 1 |
| 2 | 0 | 1 | 1 | 0 | 0 0 0 1 0 |
| 3 | 0 | 1 | 0 | 1 | 0 0 0 1 1 |
| 4 | 0 | 1 | 0 | 0 | 0 0 1 0 0 |
| 5 | 1 | 0 | 1 | 1 | 0 0 1 0 1 |
| 6 | 1 | 0 | 1 | 0 | 0 0 1 1 0 |
| 7 | 1 | 0 | 0 | 1 | 0 0 1 1 1 |
| 8 | 1 | 0 | 0 | 0 | 0 1 0 0 0 |
| 9 | 1 | 1 | 1 | 1 | 0 1 0 0 1 |
| 10 | 1 | 1 | 1 | 0 | 1 0 0 0 0 |
| 11 | 1 | 1 | 0 | 1 | 1 0 0 0 1 |
| 12 | 1 | 1 | 0 | 0 | 1 0 0 1 0 |

- The k-map for F_1

| $AB \backslash CD$ | $\bar{C}\bar{D}$ | $\bar{C}D$ | CD | $C\bar{D}$ |
|--------------------|------------------|------------|------|------------|
| $\bar{A}\bar{B}$ | 0 | 0 | 0 | 0 |
| $\bar{A}B$ | 0 | 0 | 0 | 0 |
| AB | 1 | X | X | X |
| $A\bar{B}$ | 0 | 0 | 1 | 1 |

$$\therefore F_1 = AB + AC$$

- The k-map for F_2

| $AB \backslash CD$ | $\bar{C}\bar{D}$ | $\bar{C}D$ | CD | $C\bar{D}$ |
|--------------------|------------------|------------|------|------------|
| $\bar{A}\bar{B}$ | 0 | 0 | 0 | 0 |
| $\bar{A}B$ | 0 | 0 | 0 | 0 |
| AB | 0 | X | X | X |
| $A\bar{B}$ | 1 | 1 | 0 | 0 |

$$\therefore F_2 = A\bar{B}\bar{C}$$

1) Design a combinational logic ckt which converts

1. BCD to excess 3 code

2. Excess 3 code 8, 4, -2, -1 to BCD.

3. 2, 4, 2421 ~~code~~ -2 -1

Use only 4 bit input.

- the k-map for f_1 :

| AB \ CD | 00 | 01 | 11 | 10 |
|------------------|----|----|----|----|
| $\bar{A}\bar{B}$ | 0 | 0 | 0 | 0 |
| $\bar{A}B$ | 0 | 1 | 1 | 1 |
| AB | X | X | X | X |
| $A\bar{B}$ | 1 | 1 | X | X |

$$\therefore f_1 = A + B\bar{D} + BC$$

- the k-map for f_2 :

| AB \ CD | 00 | 01 | 11 | 10 |
|------------------|----|----|----|----|
| $\bar{A}\bar{B}$ | 0 | 1 | 1 | 1 |
| $\bar{A}B$ | 1 | 0 | 0 | 0 |
| AB | X | X | X | X |
| $A\bar{B}$ | 0 | 1 | X | X |

$$\therefore f_2 = \bar{B}D + \bar{B}C + B\bar{C}\bar{D}$$

- the k-map for f_4 :

| AB \ CD | 00 | 01 | 11 | 10 |
|------------------|----|----|----|----|
| $\bar{A}\bar{B}$ | 1 | 0 | 0 | 1 |
| $\bar{A}B$ | 1 | 0 | 0 | 1 |
| AB | X | X | X | X |
| $A\bar{B}$ | 1 | 0 | X | X |

$$\therefore f_4 = \bar{D}$$

- the k-map for f_3 :

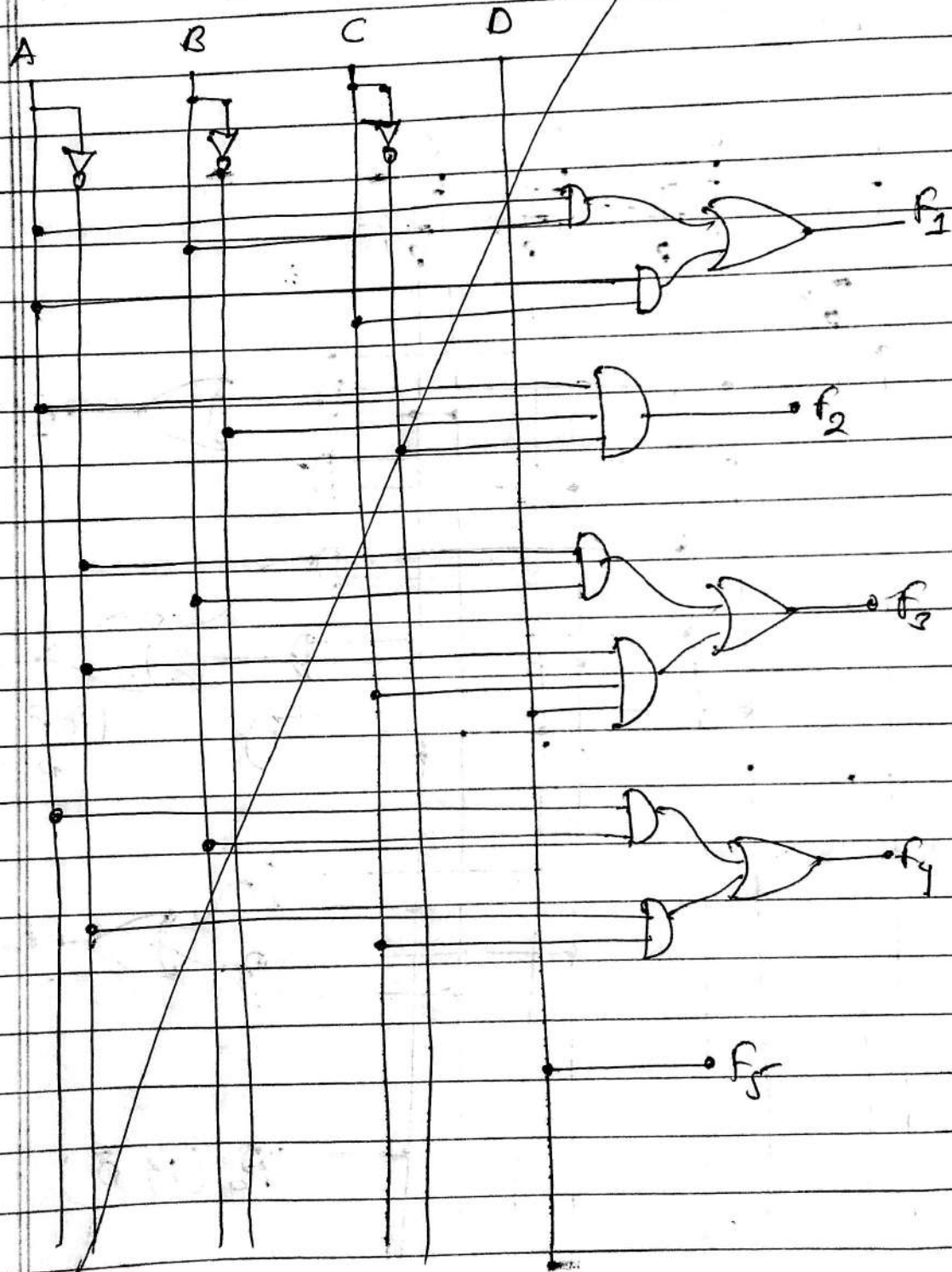
| AB \ CD | 00 | 01 | 11 | 10 |
|------------------|----|----|----|----|
| $\bar{A}\bar{B}$ | 1 | 0 | 1 | 0 |
| $\bar{A}B$ | 1 | 0 | 1 | 0 |
| AB | X | X | X | X |
| $A\bar{B}$ | 1 | 0 | X | X |

$$\therefore f_3 = \bar{C}\bar{D} + C\bar{D}$$

2.)

The combinational logic circuit is

PAGE No
DATE: / / 201



BCD Subtraction

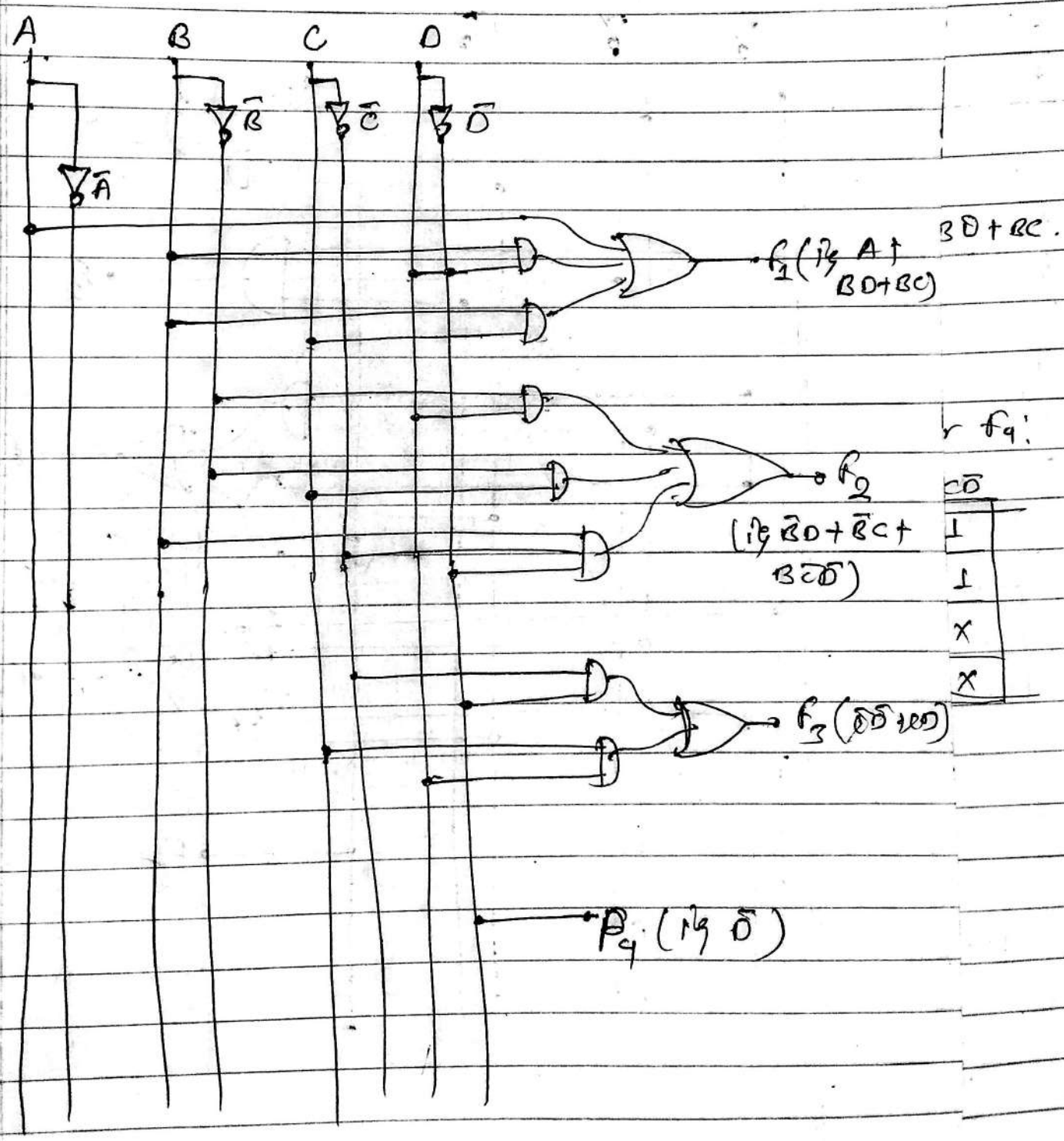
Rules:

| | |
|---|---|
| <p>End carry = 1</p> <p>→ result will be 0, add 1 to LSB</p> | <p>End carry 0</p> <p>→ result will be 0 so obtain the 1's complement</p> |
| <p>decade 1</p> <p>→ add decade</p> <p>Carry to next decade</p> | <p>decade 1</p> <p>add 1010 to that decade & ignore carry and</p> <p>decade 0</p> <p>take -ve sign.</p> |

1.

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the combinational logic ckt of above function is



Truth Table for F_4 :

| A | B | C | D | F_4 |
|---|---|---|---|-------|
| 1 | 1 | 1 | 1 | 1 |
| 1 | 1 | 1 | 0 | 1 |
| 1 | 1 | 0 | 1 | 1 |
| 1 | 1 | 0 | 0 | 0 |
| 1 | 0 | 1 | 1 | 0 |
| 1 | 0 | 1 | 0 | 0 |
| 1 | 0 | 0 | 1 | 0 |
| 1 | 0 | 0 | 0 | 0 |
| 0 | 1 | 1 | 1 | 0 |
| 0 | 1 | 1 | 0 | 0 |
| 0 | 1 | 0 | 1 | 0 |
| 0 | 1 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 | 0 |
| 0 | 0 | 1 | 0 | 0 |
| 0 | 0 | 0 | 1 | 0 |
| 0 | 0 | 0 | 0 | 0 |

Code conversion:

The Logic

Question:

- ① Design a combinational logic circuit which can
1. BCD to excess 3 code.
 2. 84-2-1 to BCD.
 3. ~~84~~ 2421 to 84-2-1.

Use only 4 bit input.

Soⁿ: $0-9 = 1001$ (is 4 bit) Ex-3 code $9+3=12$ (is 1100) $\rightarrow$ 4 bit

| BCD | | | | Ex-3 code | | | |
|-----|---|---|---|----------------|----------------|----------------|----------------|
| A | B | C | D | F ₁ | F ₂ | F ₃ | F ₄ |
| 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 |
| 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 |
| 0 | 0 | 1 | 0 | 0 | 1 | 0 | 1 |
| 0 | 0 | 1 | 1 | 0 | 1 | 1 | 0 |
| 0 | 1 | 0 | 0 | 0 | 1 | 1 | 1 |
| 0 | 1 | 0 | 1 | 1 | 0 | 0 | 0 |
| 0 | 1 | 1 | 0 | 1 | 0 | 0 | 1 |
| 0 | 1 | 1 | 1 | 1 | 0 | 1 | 0 |
| 1 | 0 | 0 | 0 | 1 | 0 | 1 | 1 |
| 1 | 0 | 0 | 1 | 1 | 1 | 0 | 0 |

for f_2 :

| $AB \backslash CD$ | $\bar{C}\bar{D}$ | $\bar{C}D$ | CD | $C\bar{D}$ |
|--------------------|------------------|------------|------|------------|
| $\bar{A}\bar{B}$ | 0 | 1 | 1 | 1 |
| $\bar{A}B$ | 1 | 0 | 0 | 0 |
| AB | X | X | X | X |
| $A\bar{B}$ | 0 | 1 | X | X |

$$\therefore F_2 = \bar{B}C + \bar{B}D + B\bar{C}\bar{D}$$

for f_3 :

| $AB \backslash CD$ | $\bar{C}\bar{D}$ | $\bar{C}D$ | CD | $C\bar{D}$ |
|--------------------|------------------|------------|------|------------|
| $\bar{A}\bar{B}$ | 1 | 0 | 1 | 0 |
| $\bar{A}B$ | 1 | 0 | 1 | 0 |
| AB | X | X | X | X |
| $A\bar{B}$ | 1 | 0 | X | X |

$$\therefore F_3 = \bar{C}\bar{D} + CD$$

Similarly for F_4 :

| $AB \backslash CD$ | $\bar{C}\bar{D}$ | $\bar{C}D$ | CD | $C\bar{D}$ |
|--------------------|------------------|------------|------|------------|
| $\bar{A}\bar{B}$ | 1 | 0 | 0 | 1 |
| $\bar{A}B$ | 1 | 0 | 0 | 1 |
| AB | X | X | X | X |
| $A\bar{B}$ | 1 | 0 | X | X |

$$\therefore F = \bar{D}$$

Design procedure of CLC:

The design of combinational ckt starts from the outline of the problem and ends in logical circuit diagram or a set of boolean function from which the logic diagram can be easily obtained. So the procedure steps are as follows.

- the problem is stated.
- the no. of variables I/p and O/p is determined.
- the input and output variables are assign by letter symbols.
- the truth table that defines the required relation betⁿ I/p and o/p is obtained.
- the simplified boolean function for each output is obtained by K-map.
- and finally logical diagram is drawn with the help of o/p boolean function.

Criteria

- a. minimum no. of gates.
- b. minimum no. of input to the gate.
- c. minimum propagation time of signal through ckt.
- d. minimum no. of interconnections.

Chapter 5

Combination Logic

A combinational ckt consist of logic gates whose o/p at any time are determined directly from the present combination of I/p.

A combinational ckt consist of I/p variables, logic gates and o/p variables. The logic gate accept signal from I/p and generates signal to the o/p.

for the 'n' I/p variable there are 2^n possible combination of binary input values.

The block diagram combination logic ckt (CLC) is:

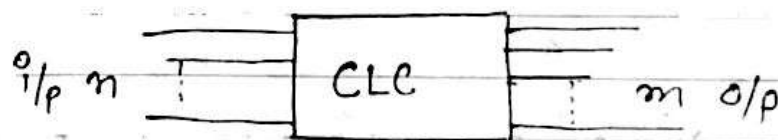


Fig: Block diagram of CLC.

for 10 circuit gates it is named MSI (Medium scale integration)

for 100 circuit gates it is named as LSI [Large scale integration]

for 1000 circuit gates it is name VLSI (Very large scale integration)

| | | |
|----|------------|------|
| 10 | 0 - 10 | MSI |
| | 10 - 100 | LSI |
| | 100 - 1000 | VLSI |

Question:

Design a circuit diagram which convert four bit BCD to excess-3 code.

| | BCD | | | | → | Ex-3 | | | |
|-----|-----|---|---|---|---|-------|-------|-------|-------|
| | A | B | C | D | | f_1 | f_2 | f_3 | f_4 |
| 0 : | 0 | 0 | 0 | 0 | → | 0 | 0 | 1 | 1 |
| 1 : | 0 | 0 | 0 | 1 | → | 0 | 1 | 0 | 0 |
| 2 : | 0 | 0 | 1 | 0 | → | 0 | 1 | 0 | 1 |
| 3 : | 0 | 0 | 1 | 1 | → | 0 | 1 | 1 | 0 |
| 4 : | 0 | 1 | 0 | 0 | → | 0 | 1 | 1 | 1 |
| 5 : | 0 | 1 | 0 | 1 | → | 1 | 0 | 0 | 0 |
| 6 : | 0 | 1 | 1 | 0 | → | 1 | 0 | 0 | 1 |
| 7 : | 0 | 1 | 1 | 1 | → | 1 | 0 | 1 | 0 |
| 8 : | 1 | 0 | 0 | 0 | → | 1 | 0 | 1 | 1 |
| 9 : | 1 | 0 | 0 | 1 | → | 1 | 0 | 0 | 0 |